Two-Timescale Secure Transmission for Movable Antenna-Enabled Multiuser Systems With Statistical Eavesdropper CSI

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***Abstract*—This paper investigates two-timescale secure transmission for a movable antenna (MA)-enabled multiuser downlink with a passive eavesdropper (Eve), where the antenna positions are optimized based on statistical channel state information (CSI), and the zero-forcing precoding with null-space artificial noise (AN) is adapted to the instantaneous CSI of the legitimate users (LUs). Specifically, the antenna positions and the long-term power allocation of the confidential signals and AN are jointly optimized to maximize the ergodic secrecy sum rate subject to the transmit power, movable-region, and minimum-distance constraints. To solve this nonconvex stochastic problem, we first transform it into a deterministic problem by approximating the ergodic wiretap rate in closed form, where the resulting expression reveals that the MAs can reduce the information leakage and increase the AN received by Eve, even for a single LU. Then, an alternating optimization algorithm with guaranteed convergence is developed for the deterministic problem, where the power allocation among the LUs follows a secrecy water-filling structure, and each antenna position is updated by a grid search followed by majorization–minimization. Simulation results demonstrate that the proposed design outperforms the benchmark schemes with less AN power and fewer antennas.**

***Index Terms*—Artificial noise, movable antenna, physical layer security, statistical CSI, two-timescale design.**

I. Introduction

PHYSICAL layer security (PLS) exploits the randomness and spatial characteristics of wireless channels to protect confidential messages against eavesdropping and thus complements upper-layer cryptography [1]–[3]. In multi-antenna systems, PLS is typically achieved by secure precoding and artificial noise (AN) [4], [5], which steer the confidential signals toward the legitimate users (LUs) and degrade the channel of the eavesdropper (Eve), respectively. Both techniques, however, rely on the spatial separability between the LUs and Eve [2], which is limited for a conventional fixed-position antenna (FPA) array when Eve is located in a direction close to that of an LU and the line-of-sight (LoS) propagation is dominant. Recently, movable antennas (MAs) [6], [7] and the closely related fluid antennas [8] have emerged as a means to enhance this separability, where adjusting the antenna positions within a given region reshapes the channel correlation among different receivers without additional radio-frequency (RF) chains [9]–[11].

For MA-enabled PLS, the transmit beamforming and the antenna positions have been jointly optimized to maximize the secrecy rate with perfect channel state information (CSI) [12] and without the instantaneous CSI of the eavesdroppers [13]. More recently, a deterministic equivalent of the ergodic secrecy rate has been derived in [14] for a multiple-input multiple-output MA system with the statistical CSI of a multi-antenna Eve. A sensing-assisted secure design has also been developed in [15], where the antenna positions improve both the estimation of the angles of departure (AoDs) of Eve and the secrecy rate. However, the designs in [12]–[14] optimize the antenna positions based on the instantaneous CSI of the legitimate receivers. Under fast fading, the antennas then have to be repositioned in each channel coherence interval (CI), which incurs nonnegligible movement delay and energy consumption. These designs also require the CSI over the entire movable region, whose acquisition entails channel measurements at many antenna positions [16], [17]. On the other hand, the design in [15] assumes LoS channels and thus does not capture the small-scale fading.

To reduce the movement and CSI acquisition overhead, the two-timescale design, which has been studied for intelligent reflecting surfaces [18], has been extended to MA systems [16], [17], where the antenna positions are optimized based on the statistical CSI, and the beamforming is designed based on the instantaneous CSI at the selected positions. The antenna positions have been optimized for a multiuser downlink with maximum-ratio transmission and zero-forcing (ZF) precoding in [16] and for a multiuser uplink with linear and successive interference cancellation receivers in [17]. However, these works focus on maximizing the ergodic sum rate without considering eavesdropping. In particular, the ergodic-rate approximations for ZF precoding adopted in [16], [17], [19] characterize only the signal power received by the intended users, and the MAs bring no rate gain for a single user [16, Remark 2]. To the best of our knowledge, the two-timescale MA design for secure transmission remains unexplored.

In this paper, we jointly optimize the antenna positions and the long-term power allocation of an MA-enabled multiuser downlink with ZF precoding and null-space AN to maximize the ergodic secrecy sum rate (ESSR). The resulting design requires neither the instantaneous CSI of Eve nor antenna movement in each CI. Since the ergodic wiretap rate depends on the random ZF directions and AN subspace induced by the channels of the LUs, we first prove that, given these channels, the information leakage and the AN received by Eve depend on orthogonal components of the wiretap channel and are thus conditionally independent. From this property, we derive a closed-form approximation of the ergodic wiretap

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rate, which captures the LoS correlation between Eve and the LUs. Unlike in the rate-oriented MA design [16], where the MAs only decorrelate the LUs, this approximation shows that the MAs also reduce the information leakage and increase the AN received by Eve, even for a single LU.

Then, we develop an alternating optimization (AO) algorithm with guaranteed convergence, whose complexity does not depend on the number of channel realizations. For given antenna positions and AN power, the optimal power allocation among the LUs follows a secrecy water-filling structure, which switches off the LUs whose effective channel gains do not exceed that of Eve. For a given power allocation, each antenna position is updated by a grid search followed by majorization–minimization (MM) refinement, where a rank-one structure of the objective function reduces the cost of evaluating each grid point. Simulation results show that the proposed design outperforms the rate-oriented MA design, antenna selection (AS) from twice as many antennas, and both compact and sparse FPA arrays, while requiring less AN power. With only four MAs, it even outperforms a compact FPA array with 12 antennas. The secrecy gain also persists for a single LU, against an Eve that treats the inter-user interference as noise, under moderate AoD errors, and for independent AoDs of Eve and the LUs.

Notations: Scalars, vectors, and matrices are denoted by italic, bold lower-case, and bold upper-case letters, respectively. $\mathbb{C}^{a\times b}$ and $\mathbb{R}^{a\times b}$ denote the sets of $a\times b$ complex and real matrices, respectively. $(\cdot)^{\mathrm{T}}$, $(\cdot)^*$, and $(\cdot)^{\mathrm{H}}$ denote the transpose, conjugate, and conjugate transpose, and $|\cdot|$, $\|\cdot\|$, and $\mathrm{tr}(\cdot)$ the modulus (or the cardinality of a set), Euclidean norm, and trace, respectively. $[\mathbf{x}]_i$ and $[\mathbf{X}]_{i,j}$ denote the i-th entry of $\mathbf{x}$ and the (i,j)-th entry of $\mathbf{X}$, $\mathrm{diag}(x_1,\ldots,x_M)$ the diagonal matrix with diagonal entries $x_1,\ldots,x_M$, and $\mathbf{X}^{1/2}$ the entrywise square root of a nonnegative diagonal matrix $\mathbf{X}$. $\mathbf{I}_N$ denotes the $N\times N$ identity matrix, $\mathbf{e}_m$ the m-th column of an identity matrix, and $\mathcal{CN}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ the proper complex Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. $\mathbf{X}\succeq\mathbf{0}$ ($\mathbf{X}\succ\mathbf{0}$) means that $\mathbf{X}$ is positive semidefinite (definite), and $\mathbf{X}_1\succeq\mathbf{X}_2$ means $\mathbf{X}_1-\mathbf{X}_2\succeq\mathbf{0}$. $j\triangleq\sqrt{-1}$, $\mathbb{E}\{\cdot\}$ and $\mathrm{Var}\{\cdot\}$ denote the expectation and variance, $\mathrm{Re}\{\cdot\}$ the real part, $\nabla_{\mathbf{x}}f$ the gradient of f with respect to $\mathbf{x}$, $[x]^+\triangleq\max\{x,0\}$, $\emptyset$ the empty set, $\setminus$ the set difference, and $\mathcal{O}(\cdot)$ and $o(\cdot)$ the big-O and little-o notations, respectively.

II. System Model and Problem Formulation

A. MA-Enabled Secure Multiuser Downlink System

As illustrated in Fig. 1, we consider the downlink of a multiuser system, where a base station (BS) equipped with N MAs serves M single-antenna LUs in the presence of a single-antenna passive Eve. Let $\mathcal{M}\triangleq\{1,\ldots,M\}$ and $\mathcal{N}\triangleq\{1,\ldots,N\}$ denote the index sets of the LUs and the MAs, respectively, where $N\geq M+1$ is required to transmit AN in the null space of the channels of the LUs. The MAs are connected to the RF chains via flexible cables and can move within a two-dimensional square region $\mathcal{C}\triangleq[-A/2,A/2]\times[-A/2,A/2]$ of side length A [6], [7]. In the local Cartesian coordinate system of $\mathcal{C}$, the position of the n-th MA is denoted by $\mathbf{t}_n\triangleq[x_n,y_n]^{\mathrm{T}}\in\mathcal{C}$, and the positions of

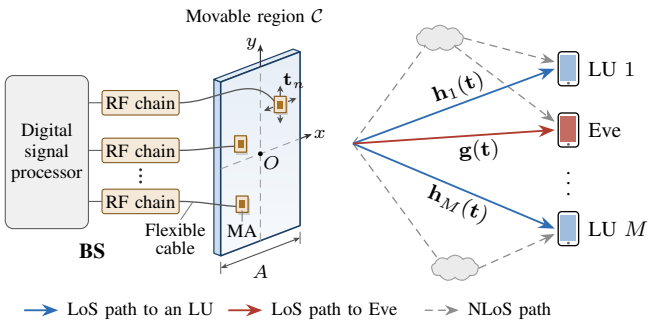


Fig. 1. An MA-enabled multiuser downlink system with a passive Eve.

all the MAs are collected in $\mathbf{t}\triangleq[\mathbf{t}_1^{\mathrm{T}},\ldots,\mathbf{t}_N^{\mathrm{T}}]^{\mathrm{T}}\in\mathbb{R}^{2N\times1}$. To mitigate mutual coupling, any two MAs must be separated by at least a minimum distance $D_{\min}$, and the residual coupling is neglected.

Because the size of $\mathcal{C}$ is much smaller than the propagation distance, moving the MAs changes only the phases of the LoS paths, whereas their AoDs and amplitudes remain unchanged [7]. Let $\theta_m\in[-\pi/2,\pi/2]$ and $\phi_m\in[-\pi/2,\pi/2]$ denote the elevation and azimuth AoDs of the LoS path from the BS to the m-th LU, respectively, and define $\mathbf{a}_m\triangleq[\cos\theta_m\sin\phi_m,\ \sin\theta_m]^{\mathrm{T}}$. The propagation distance of the LoS path from the n-th MA then differs from that from the origin of $\mathcal{C}$ by $\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_m$, so that the LoS component of the channel from the BS to the m-th LU is given by [16]

$$\bar{\mathbf{h}}_m(\mathbf{t})\triangleq\left[e^{j\frac{2\pi}{\lambda}\mathbf{t}_1^{\mathrm{T}}\mathbf{a}_m},\ldots,e^{j\frac{2\pi}{\lambda}\mathbf{t}_N^{\mathrm{T}}\mathbf{a}_m}\right]^{\mathrm{T}}\in\mathbb{C}^{N\times1},\qquad(1)$$

where λ is the carrier wavelength. As in [16], [17], the channel from the BS to the m-th LU follows Rician fading and is given by

$$\mathbf{h}_m(\mathbf{t})=\sqrt{\frac{\kappa_m\beta_m}{\kappa_m+1}}\,\bar{\mathbf{h}}_m(\mathbf{t})+\sqrt{\frac{\beta_m}{\kappa_m+1}}\,\tilde{\mathbf{h}}_m,\qquad(2)$$

where β_m is the large-scale fading coefficient, $\kappa_m\geq0$ is the Rician factor, and $\tilde{\mathbf{h}}_m\sim\mathcal{CN}(\mathbf{0},\mathbf{I}_N)$ is the non-LoS (NLoS) component.[1] Similarly, the channel from the BS to Eve is given by

$$\mathbf{g}(\mathbf{t})=\sqrt{\frac{\kappa_{\mathrm{e}}\beta_{\mathrm{e}}}{\kappa_{\mathrm{e}}+1}}\,\bar{\mathbf{g}}(\mathbf{t})+\sqrt{\frac{\beta_{\mathrm{e}}}{\kappa_{\mathrm{e}}+1}}\,\tilde{\mathbf{g}},\qquad(3)$$

where β_{e}, κ_{e}, and $\tilde{\mathbf{g}}\sim\mathcal{CN}(\mathbf{0},\mathbf{I}_N)$ are defined similarly to those in (2), and $\bar{\mathbf{g}}(\mathbf{t})$ is given by (1) with $\mathbf{a}_m$ replaced by $\mathbf{a}_{\mathrm{e}}\triangleq[\cos\theta_{\mathrm{e}}\sin\phi_{\mathrm{e}},\ \sin\theta_{\mathrm{e}}]^{\mathrm{T}}$ for the elevation and azimuth AoDs θ_{e} and ϕ_{e} of Eve. The NLoS components $\tilde{\mathbf{h}}_1,\ldots,\tilde{\mathbf{h}}_M$ and $\tilde{\mathbf{g}}$ are mutually independent. Let $\mathbf{H}(\mathbf{t})\triangleq[\mathbf{h}_1(\mathbf{t}),\ldots,\mathbf{h}_M(\mathbf{t})]\in\mathbb{C}^{N\times M}$ denote the channel matrix from the BS to the LUs, and let $\bar{\mathbf{H}}(\mathbf{t})\triangleq[\bar{\mathbf{h}}_1(\mathbf{t}),\ldots,\bar{\mathbf{h}}_M(\mathbf{t})]$ and $\tilde{\mathbf{H}}\triangleq[\tilde{\mathbf{h}}_1,\ldots,\tilde{\mathbf{h}}_M]$ collect its LoS and NLoS components, respectively. The Rician factors of the LUs are collected in $\mathbf{K}\triangleq\mathrm{diag}(\kappa_1,\ldots,\kappa_M)$.

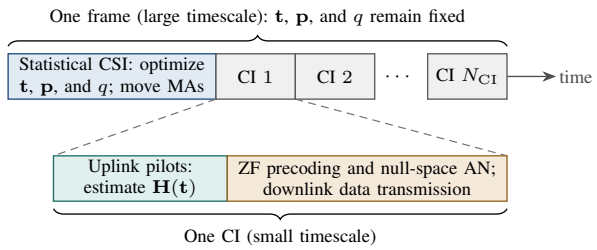


Fig. 2. Two-timescale transmission protocol (CI: coherence interval).

B. Two-Timescale Transmission Protocol

Unlike the instantaneous CSI over the entire region $\mathcal{C}$, the statistical CSI, i.e., $\{\beta_m, \kappa_m, \theta_m, \phi_m\}_{m\in\mathcal{M}}$ of the LUs and $\{\beta_\mathrm{e}, \kappa_\mathrm{e}, \theta_\mathrm{e}, \phi_\mathrm{e}\}$ of Eve, is determined by the large-scale fading and the LoS geometry, which vary much more slowly than the small-scale fading. Following [13], we assume that the statistical CSI of Eve is known at the BS.[2]

Accordingly, we adopt the two-timescale protocol in Fig. 2, which divides the time into frames of $N_\mathrm{CI} \gg 1$ CIs, over which the statistical CSI remains constant. At the beginning of each frame, the BS optimizes the antenna positions $\mathbf{t}$ and the long-term power allocation $(\mathbf{p}, q)$ specified in Section III-A based on the statistical CSI, and moves the MAs accordingly. In each CI, the BS then estimates $\mathbf{H}(\mathbf{t})$ from uplink pilots by exploiting the channel reciprocity of time-division duplexing, and designs the precoding and AN. For ease of exposition, $\mathbf{H}(\mathbf{t})$ is assumed to be perfectly estimated, whereas the NLoS component $\tilde{\mathbf{g}}$ of Eve is never available at the BS. As a result, the MAs move only once per frame, and the CSI of the LUs is needed only at the current antenna positions. The movement delay, which is amortized over the N_CI CIs of a frame, and the pilot overhead are not included in the ergodic rates defined below.

C. Signal Model and Ergodic Secrecy Rate

In each CI, the BS transmits the confidential symbols $s_m \sim \mathcal{CN}(0,1)$, $m \in \mathcal{M}$, together with AN, and the transmit signal is given by

$$\mathbf{x} = \sum_{m=1}^{M} \mathbf{w}_m s_m + \mathbf{z}, \tag{4}$$

where $\mathbf{w}_m \in \mathbb{C}^{N\times 1}$ is the precoding vector for the m-th LU, and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma})$ is the AN with covariance matrix $\mathbf{\Sigma} \succeq \mathbf{0}$. The symbols $s_1, \ldots, s_M$ and the AN $\mathbf{z}$ are mutually independent. The transmit power satisfies $\sum_{m=1}^{M} \|\mathbf{w}_m\|^2 + \mathrm{tr}(\mathbf{\Sigma}) \le P_\mathrm{tot}$, where P_tot is the maximum transmit power of the BS. Since $\mathbf{w}_m$ and $\mathbf{\Sigma}$ depend on the instantaneous CSI of the LUs, we write them as $\mathbf{w}_m(\mathbf{H}(\mathbf{t}))$ and $\mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))$.

The signal received at the m-th LU is $y_m = \mathbf{h}_m^\mathrm{H}(\mathbf{t})\mathbf{x} + n_m$, where $n_m \sim \mathcal{CN}(0, \sigma_m^2)$ is the additive white Gaussian noise. The resulting signal-to-interference-plus-noise ratio (SINR) of the m-th LU is

$$\gamma_m = \frac{|\mathbf{h}_m^\mathrm{H}(\mathbf{t})\mathbf{w}_m|^2}{\sum_{i=1, i\neq m}^{M} |\mathbf{h}_m^\mathrm{H}(\mathbf{t})\mathbf{w}_i|^2 + \mathbf{h}_m^\mathrm{H}(\mathbf{t})\mathbf{\Sigma}\mathbf{h}_m(\mathbf{t}) + \sigma_m^2}. \tag{5}$$

The signal received at Eve is $y_\mathrm{e} = \mathbf{g}^\mathrm{H}(\mathbf{t})\mathbf{x} + n_\mathrm{e}$ with $n_\mathrm{e} \sim \mathcal{CN}(0, \sigma_\mathrm{e}^2)$. Since the BS does not know the receiver of a passive Eve, we consider the worst case where Eve can cancel the signals of the other LUs when intercepting s_m, e.g., via multiuser detection [20]. This worst-case model provides a secrecy guarantee that does not depend on the decoding capability of Eve. Under this model, the SINR of Eve for s_m is given by

$$\gamma_{\mathrm{e},m} = \frac{|\mathbf{g}^\mathrm{H}(\mathbf{t})\mathbf{w}_m|^2}{\mathbf{g}^\mathrm{H}(\mathbf{t})\mathbf{\Sigma}\mathbf{g}(\mathbf{t}) + \sigma_\mathrm{e}^2}, \tag{6}$$

which upper-bounds that of an Eve treating the signals of the other LUs as interference.

Since the instantaneous CSI of Eve is unknown, the confidential message of each LU is encoded over the CIs of a frame, which experience independent realizations of the NLoS components. Under this coding scheme, an achievable ergodic secrecy rate of the m-th LU is given by [20], [21]

$$R_m^\mathrm{sec} \triangleq \left[\bar{R}_m - \bar{R}_{\mathrm{e},m}\right]^+, \tag{7}$$

where

$$\bar{R}_m \triangleq \mathbb{E}\{\log_2(1+\gamma_m)\}, \quad \bar{R}_{\mathrm{e},m} \triangleq \mathbb{E}\{\log_2(1+\gamma_{\mathrm{e},m})\} \tag{8}$$

denote the ergodic rate of the m-th LU and the ergodic wiretap rate of Eve for s_m, respectively, with the expectations taken over $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{g}}$ for a given $\mathbf{t}$. The ESSR is then given by $\sum_{m=1}^{M} R_m^\mathrm{sec}$.

D. Problem Formulation

We aim to maximize the ESSR by jointly optimizing the antenna positions and the small-timescale policy $\mathcal{W} \triangleq \{\mathbf{w}_1(\mathbf{H}(\mathbf{t})), \ldots, \mathbf{w}_M(\mathbf{H}(\mathbf{t})), \mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))\}$, which maps the instantaneous CSI of the LUs to the precoders and the AN covariance matrix. The resulting problem is formulated as

$$\begin{aligned}
(\mathrm{P1}):\ \max_{\mathbf{t},\,\mathcal{W}}\ & \sum_{m=1}^{M} R_m^\mathrm{sec} && \text{(9a)}\\
\text{s.t.}\ & \sum_{m=1}^{M} \left\|\mathbf{w}_m(\mathbf{H}(\mathbf{t}))\right\|^2 + \mathrm{tr}\big(\mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))\big) \le P_\mathrm{tot},\\
& \mathbf{\Sigma}(\mathbf{H}(\mathbf{t})) \succeq \mathbf{0},\ \forall \mathbf{H}(\mathbf{t}), && \text{(9b)}\\
& \mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, && \text{(9c)}\\
& \|\mathbf{t}_n - \mathbf{t}_i\| \ge D_\mathrm{min},\ \forall n, i \in \mathcal{N},\ n \neq i. && \text{(9d)}
\end{aligned}$$

Problem (P1) is a nonconvex stochastic optimization problem, which is difficult to solve for two reasons. First, the ergodic rates in (8) have no closed-form expressions, and the policy $\mathcal{W}$ has to be designed for every realization of $\mathbf{H}(\mathbf{t})$. Second, since the phases of the LoS components in (1) vary with $\mathbf{t}$, the objective function is nonconcave in $\mathbf{t}$, and the minimum-distance constraints in (9d) are nonconvex. To address the first difficulty, Section III restricts $\mathcal{W}$ to ZF precoding

[1] As in [16], [17], the NLoS components at different antennas are assumed to be independent and identically distributed (i.i.d.), which is an idealized rich-scattering model adopted for analytical tractability. In practice, the minimum distance $D_\mathrm{min} \ge \lambda/2$ reduces, but does not eliminate, the spatial correlation.

[2] The statistical CSI of Eve can be obtained, e.g., from prior knowledge of its location, by sensing its direction [15], or from its historical signals if it is an untrusted node.

and null-space AN with a long-term power allocation and approximates the ergodic rates in closed form, which yields a deterministic problem in the long-term variables only. To address the second difficulty, Section IV develops an AO algorithm for this problem.

III. Two-Timescale ZF-AN Design and Ergodic Secrecy Rate Analysis

A. Small-Timescale ZF Precoding and Null-Space AN

To avoid designing $\mathcal{W}$ for every realization of $\mathbf{H}(\mathbf{t})$, we restrict the small-timescale policy to ZF precoding and null-space AN with a long-term power allocation. In each CI, the precoder of the m-th LU is $\mathbf{w}_m(\mathbf{H}(\mathbf{t})) = \sqrt{p_m}\,\bar{\mathbf{w}}_m$, where $p_m \geq 0$ denotes the power allocated to the m-th LU, and

$$\bar{\mathbf{w}}_m \triangleq \frac{\mathbf{H}(\mathbf{t})\big(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{e}_m}{\big\|\mathbf{H}(\mathbf{t})\big(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{e}_m\big\|} \tag{10}$$

is the normalized ZF precoding vector of the m-th LU, which satisfies $\mathbf{h}_i^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m = 0,\ \forall i \neq m$. The AN is isotropically distributed in the null space of $\mathbf{H}^{\mathrm{H}}(\mathbf{t})$ [5], [20], i.e., $\mathbf{\Sigma}(\mathbf{H}(\mathbf{t})) = q\,\mathbf{P}^{\perp}(\mathbf{t})$, where $q \geq 0$ denotes the AN power per dimension, and

$$\mathbf{P}^{\perp}(\mathbf{t}) \triangleq \mathbf{I}_N - \mathbf{H}(\mathbf{t})\big(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{H}^{\mathrm{H}}(\mathbf{t}) \tag{11}$$

is the orthogonal projection onto the null space of $\mathbf{H}^{\mathrm{H}}(\mathbf{t})$. Since $\mathbf{H}(\mathbf{t})$ has full column rank with probability one, we have $\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{0}$ and $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$. The AN therefore causes no interference to the LUs, and the transmit power in each CI is $\sum_{m=1}^{M} p_m + (N-M)q$.

We refer to $\mathbf{p} \triangleq [p_1, \ldots, p_M]^{\mathrm{T}}$ and q as the long-term power allocation, which is optimized based on the statistical CSI and remains fixed within a frame, whereas $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are updated in each CI according to $\mathbf{H}(\mathbf{t})$. With this design, the SINRs in (5) and (6) reduce to

$$\gamma_m = \frac{p_m|\mathbf{h}_m^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2}{\sigma_m^2} = \frac{p_m}{\sigma_m^2\big[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}\big]_{m,m}}, \tag{12}$$

$$\gamma_{\mathrm{e},m} = \frac{p_m|\mathbf{g}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2}{q\,\mathbf{g}^{\mathrm{H}}(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t}) + \sigma_{\mathrm{e}}^2}, \tag{13}$$

where the second equality in (12) follows from $|\mathbf{h}_m^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2 = 1/[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}$ by (10).

The ZF-AN design requires only the instantaneous CSI of the LUs and involves no optimization in each CI, because $\mathbf{p}$ and q depend only on the statistical CSI. The equal power allocation in [16] and the information–AN power splitting with equal power per LU are special cases of $(\mathbf{p}, q)$. With the ZF-AN design, (P1) is restricted to

$$\begin{aligned}
(\mathrm{P2}):\ \max_{\mathbf{t},\mathbf{p},q}\ & \sum_{m=1}^{M} R_m^{\mathrm{sec}}(\mathbf{t}, p_m, q) && \text{(14a)}\\
\text{s.t.}\ & \sum_{m=1}^{M} p_m + (N-M)q \leq P_{\mathrm{tot}}, && \text{(14b)}\\
& p_m \geq 0,\ \forall m \in \mathcal{M}, \quad q \geq 0, && \text{(14c)}\\
& \mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, && \text{(14d)}\\
& \|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min},\ \forall n, i \in \mathcal{N},\ n \neq i, && \text{(14e)}
\end{aligned}$$

where $R_m^{\mathrm{sec}}(\mathbf{t}, p_m, q)$ denotes (7) with the SINRs in (12) and (13), and (14b) and (14c) guarantee (9b) in every CI. Since the ZF-AN design is a particular policy $\mathcal{W}$, (P2) is a restriction of (P1): every feasible solution of (P2) is feasible for (P1). However, (P2) remains intractable because the ergodic rates in (14a) have no closed-form expressions. The ergodic rates of the LUs and Eve are therefore approximated in Sections III-B and III-C, respectively.

B. Ergodic Rate of the LUs

Define $\mathbf{B} \triangleq \mathrm{diag}\big(\frac{\beta_1}{\kappa_1+1}, \ldots, \frac{\beta_M}{\kappa_M+1}\big)$ and $\mathbf{L}(\mathbf{t}) \triangleq \bar{\mathbf{H}}(\mathbf{t})\mathbf{K}^{1/2}$. From (2), the channel matrix of the LUs can be written as

$$\mathbf{H}(\mathbf{t}) = \breve{\mathbf{H}}(\mathbf{t})\mathbf{B}^{1/2}, \qquad \breve{\mathbf{H}}(\mathbf{t}) \triangleq \mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}, \tag{15}$$

where $\breve{\mathbf{H}}(\mathbf{t})$ is the normalized channel matrix, whose LoS component is weighted by the square roots of the Rician factors. We further define

$$\mathbf{\Phi}(\mathbf{t}) \triangleq \mathbf{L}^{\mathrm{H}}(\mathbf{t})\mathbf{L}(\mathbf{t}) + N\mathbf{I}_M, \qquad \mathbf{\Omega}(\mathbf{t}) \triangleq \mathbf{\Phi}^{-1}(\mathbf{t}), \tag{16}$$

where $\mathbf{\Phi}(\mathbf{t}) = \mathbb{E}\{\breve{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\breve{\mathbf{H}}(\mathbf{t})\}$ is the mean of the Gram matrix of the normalized channels of the LUs.

Since $\log_2(1 + p_m/(\sigma_m^2 z))$ is convex in $z > 0$, applying Jensen's inequality to (12) and using (15) yield

$$\bar{R}_m \geq \log_2\left(1 + \frac{p_m[\mathbf{B}]_{m,m}}{\sigma_m^2\mathbb{E}\big\{[(\breve{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\breve{\mathbf{H}}(\mathbf{t}))^{-1}]_{m,m}\big\}}\right). \tag{17}$$

Following [19], we approximate the Gram matrix $\breve{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\breve{\mathbf{H}}(\mathbf{t})$ by a central Wishart matrix with N degrees of freedom and the same mean $\mathbf{\Phi}(\mathbf{t})$, whose inverse has the mean $N\mathbf{\Omega}(\mathbf{t})/(N-M)$. Substituting this mean into (17) yields

$$\bar{R}_m \approx \tilde{R}_m(\mathbf{t}, p_m) \triangleq \log_2\big(1 + \eta_m(\mathbf{t})\,p_m\big), \tag{18}$$

where $\eta_m(\mathbf{t}) \triangleq \zeta_m(N-M)/(N\omega_m(\mathbf{t}))$ is the effective channel gain of the m-th LU, $\zeta_m \triangleq \beta_m/((\kappa_m+1)\sigma_m^2)$, and $\omega_m(\mathbf{t}) \triangleq [\mathbf{\Omega}(\mathbf{t})]_{m,m}$. The approximation in (18) is equivalent to that in [16, Sec. IV-A]. For Rayleigh fading, the central Wishart model is exact, and (18) reduces to the well-known lower bound $\log_2(1 + \zeta_m p_m(N-M))$ [22]. In the LoS-dominant regime, (18) is conservative, as shown in Section V-B. The antenna positions affect $\tilde{R}_m(\mathbf{t}, p_m)$ only through $\omega_m(\mathbf{t})$. Since $\mathbf{\Phi}(\mathbf{t}) \succeq N\mathbf{I}_M$ and $[\mathbf{\Omega}(\mathbf{t})]_{m,m} \geq 1/[\mathbf{\Phi}(\mathbf{t})]_{m,m} = 1/(N(\kappa_m+1))$ [23], we have

$$\zeta_m(N-M) \leq \eta_m(\mathbf{t}) \leq \zeta_m(N-M)(\kappa_m+1), \tag{19}$$

where the upper bound is achieved if $\bar{\mathbf{h}}_m^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{h}}_i(\mathbf{t}) = 0$ for all $i \neq m$ with $\kappa_i > 0$. The ratio $1/(N\omega_m(\mathbf{t})) \in [1, \kappa_m + 1]$ is thus the LoS gain of the m-th LU that survives the ZF precoding, which the antenna positions can improve by decorrelating the LoS components of the LUs.

C. Ergodic Wiretap Rate of Eve

Unlike γ_m in (12), which depends on $\mathbf{H}(\mathbf{t})$ only through $[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}$, the SINR of Eve in (13) depends on the random ZF direction $\bar{\mathbf{w}}_m$ and AN subspace, both determined by $\mathbf{H}(\mathbf{t})$. This dependence is the main difficulty in deriving

$\bar{R}_{\mathrm{e},m}$, and we therefore first characterize the information leakage and the AN received by Eve for a given $\mathbf{H}(\mathbf{t})$.

Lemma 1: Conditioned on $\mathbf{H}(\mathbf{t})$, the information leakage $X_m \triangleq |\mathbf{g}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2$ and the AN power $Y \triangleq \mathbf{g}^{\mathrm{H}}(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t})$ received by Eve are independent, and their conditional means and variances are given by

$$\begin{aligned} \mathbb{E}\{X_m|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_{\mathrm{e}}\big(\kappa_{\mathrm{e}}\bar{X}_m + 1\big), \\ \mathrm{Var}\{X_m|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_{\mathrm{e}}^2\big(2\kappa_{\mathrm{e}}\bar{X}_m + 1\big), \end{aligned} \tag{20}$$

$$\begin{aligned} \mathbb{E}\{Y|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_{\mathrm{e}}\big(\kappa_{\mathrm{e}}\bar{Y} + N - M\big), \\ \mathrm{Var}\{Y|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_{\mathrm{e}}^2\big(2\kappa_{\mathrm{e}}\bar{Y} + N - M\big), \end{aligned} \tag{21}$$

where $\tilde{\beta}_{\mathrm{e}} \triangleq \beta_{\mathrm{e}}/(\kappa_{\mathrm{e}}+1)$, and $\bar{X}_m \triangleq |\bar{\mathbf{g}}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{w}}_m|^2$ and $\bar{Y} \triangleq \bar{\mathbf{g}}^{\mathrm{H}}(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$ are the LoS counterparts of X_m and Y.

Proof: See Appendix A. ∎

By Lemma 1, the information leakage for given channels of the LUs is determined only by the component of the wiretap channel along the ZF direction, whereas the AN received by Eve is determined only by its component in the null space of $\mathbf{H}^{\mathrm{H}}(\mathbf{t})$. The conditional relative variance of X_m, $(2\kappa_{\mathrm{e}}\bar{X}_m + 1)/(\kappa_{\mathrm{e}}\bar{X}_m + 1)^2$, is small when the LoS leakage $\kappa_{\mathrm{e}}\bar{X}_m$ is strong, and that of Y, $(2\kappa_{\mathrm{e}}\bar{Y} + N - M)/(\kappa_{\mathrm{e}}\bar{Y} + N - M)^2$, is small when $N - M$ or $\kappa_{\mathrm{e}}\bar{Y}$ is large. However, $\bar{X}_m$ and $\bar{Y}$ also vary with $\tilde{\mathbf{H}}$, e.g., $\mathrm{Var}\{X_m\} = \tilde{\beta}_{\mathrm{e}}^2(2\kappa_{\mathrm{e}}\mathbb{E}\{\bar{X}_m\} + 1 + \kappa_{\mathrm{e}}^2\mathrm{Var}\{\bar{X}_m\})$, so these conditions alone do not ensure the concentration of X_m and Y.

Nevertheless, as in [19, Lemma 1], we approximate $\bar{R}_{\mathrm{e},m} = \mathbb{E}\{\log_2(1 + p_m X_m/(qY + \sigma_{\mathrm{e}}^2))\}$ by $\log_2(1 + p_m\mathbb{E}\{X_m\}/(q\mathbb{E}\{Y\} + \sigma_{\mathrm{e}}^2))$, which is accurate when X_m and Y concentrate around their means and is asymptotically exact as $\kappa_{\mathrm{e}}, \kappa_m \to \infty$, $\forall m \in \mathcal{M}$, provided that the LoS components of the LUs are linearly independent. Because the rate is concave in X_m and convex in Y, this approximation is neither an upper nor a lower bound; its accuracy is evaluated in Section V-B. Using Lemma 1 and taking the expectation over $\tilde{\mathbf{H}}$, we obtain

$$\bar{R}_{\mathrm{e},m} \approx \log_2\left(1 + \frac{p_m\big(\kappa_{\mathrm{e}}\mathbb{E}\{\bar{X}_m\} + 1\big)}{q\big(\kappa_{\mathrm{e}}\mathbb{E}\{\bar{Y}\} + N - M\big) + \bar{\sigma}_{\mathrm{e}}^2}\right), \tag{22}$$

where $\bar{\sigma}_{\mathrm{e}}^2 \triangleq \sigma_{\mathrm{e}}^2/\tilde{\beta}_{\mathrm{e}}$ is the normalized noise power of Eve.

The expectations $\mathbb{E}\{\bar{X}_m\}$ and $\mathbb{E}\{\bar{Y}\}$ in (22), which are the average LoS energies of Eve along the ZF direction of the m-th LU and in the AN subspace, respectively, still have no closed-form expressions. Since $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are invariant to the scaling of the columns of $\mathbf{H}(\mathbf{t})$, they depend on $\mathbf{H}(\mathbf{t})$ only through $\breve{\mathbf{H}}(\mathbf{t})$ in (15). As derived in Appendix B, replacing the inverse of the random Gram matrix $\breve{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\breve{\mathbf{H}}(\mathbf{t})$ with the inverse of its mean $\mathbf{\Phi}(\mathbf{t})$, where the NLoS contribution to the AN subspace is calibrated by the identity $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$, yields $\mathbb{E}\{\bar{X}_m\} \approx c_m(\mathbf{t})$ and $\mathbb{E}\{\bar{Y}\} \approx d(\mathbf{t})$ with

$$c_m(\mathbf{t}) \triangleq \frac{\big|[\mathbf{\Omega}(\mathbf{t})\mathbf{v}(\mathbf{t})]_m\big|^2 + N[\mathbf{\Omega}^2(\mathbf{t})]_{m,m}}{\omega_m(\mathbf{t})}, \tag{23}$$

$$d(\mathbf{t}) \triangleq \frac{(N - M)\big(N - \mathbf{v}^{\mathrm{H}}(\mathbf{t})\mathbf{\Omega}(\mathbf{t})\mathbf{v}(\mathbf{t})\big)}{N - M + N\mathrm{tr}(\mathbf{\Omega}(\mathbf{t}))}, \tag{24}$$

where $\mathbf{v}(\mathbf{t}) \triangleq \mathbf{L}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t}) = \mathbf{K}^{1/2}\bar{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$. Substituting (23) and (24) into (22), we approximate the ergodic wiretap rate of Eve for s_m by

$$\bar{R}_{\mathrm{e},m} \approx \tilde{R}_{\mathrm{e},m}(\mathbf{t}, p_m, q) \triangleq \log_2\left(1 + \frac{\xi_m(\mathbf{t})\, p_m}{q\,\psi(\mathbf{t}) + \bar{\sigma}_{\mathrm{e}}^2}\right), \tag{25}$$

where $\xi_m(\mathbf{t}) \triangleq \kappa_{\mathrm{e}}c_m(\mathbf{t}) + 1$ and $\psi(\mathbf{t}) \triangleq \kappa_{\mathrm{e}}d(\mathbf{t}) + N - M$ represent the normalized information leakage for the m-th LU and the normalized AN power received by Eve, respectively. The approximations $c_m(\mathbf{t})$ and $d(\mathbf{t})$ have the following properties.

Proposition 1: The quantities $c_m(\mathbf{t})$, $d(\mathbf{t})$, $\mathbb{E}\{\bar{X}_m\}$, and $\mathbb{E}\{\bar{Y}\}$ lie in $[0, N]$. Moreover, $c_m(\mathbf{t}) = \mathbb{E}\{\bar{X}_m\}$ and $d(\mathbf{t}) = \mathbb{E}\{\bar{Y}\}$ if $\kappa_m = 0$, $\forall m \in \mathcal{M}$. For linearly independent LoS components of the LUs, $c_m(\mathbf{t}) - \mathbb{E}\{\bar{X}_m\}$ and $d(\mathbf{t}) - \mathbb{E}\{\bar{Y}\}$ vanish as $\mathbf{K} = \kappa\mathbf{K}_0$ with $\kappa \to \infty$ for any diagonal $\mathbf{K}_0 \succ \mathbf{0}$.

Proof: See Appendix B. ∎

Unlike the approximations for ZF precoding in [16], [19], which involve only the diagonal entries of $\mathbf{\Omega}(\mathbf{t})$, (25) captures the LoS correlation between Eve and the LUs via $\mathbf{v}(\mathbf{t})$.

D. Approximate ESSR Maximization Problem

Combining (18) and (25), we approximate the ergodic secrecy rate of the m-th LU by

$$R_m^{\mathrm{sec}} \approx \tilde{R}_m^{\mathrm{sec}}(\mathbf{t}, p_m, q) \triangleq \big[\tilde{R}_m(\mathbf{t}, p_m) - \tilde{R}_{\mathrm{e},m}(\mathbf{t}, p_m, q)\big]^+, \tag{26}$$

which depends only on the statistical CSI and can thus be optimized on the large timescale. Remarks 1–3 show how the antenna positions affect the approximate ESSR.

Remark 1 (Secrecy Mechanisms of the MAs): From (18), (23), and (24), the antenna positions affect the approximate ESSR via $\omega_m(\mathbf{t})$, $c_m(\mathbf{t})$, and $d(\mathbf{t})$, i.e., via the LoS array correlations $[\bar{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{H}}(\mathbf{t})]_{i,j} = \sum_{n=1}^{N} e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}(\mathbf{a}_j - \mathbf{a}_i)}$ among the LUs and $[\bar{\mathbf{H}}^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})]_i = \sum_{n=1}^{N} e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}(\mathbf{a}_{\mathrm{e}} - \mathbf{a}_i)}$ between Eve and the LUs. A weaker LoS correlation between the m-th LU and the other LUs reduces $\omega_m(\mathbf{t})$ and thus increases the ZF gain in (19). This decorrelation among the LUs is the only mechanism exploited by the rate-oriented MA design [16]. Secure transmission offers two additional mechanisms: a smaller $c_m(\mathbf{t})$ reduces the information leakage, and a larger $d(\mathbf{t})$ increases the AN received by Eve. In particular, if $\mathbf{v}(\mathbf{t}) = \mathbf{0}$, i.e., the LoS component of Eve is orthogonal to those of all the LUs, then $c_m(\mathbf{t})$ reduces to its minimum $N[\mathbf{\Omega}^2(\mathbf{t})]_{m,m}/\omega_m(\mathbf{t})$ caused by the NLoS components, and $d(\mathbf{t})$ achieves its maximum $N(N - M)/(N - M + N\mathrm{tr}(\mathbf{\Omega}(\mathbf{t})))$ for a given $\mathbf{\Phi}(\mathbf{t})$. By decorrelating the LoS component of Eve from those of the LUs, the MAs therefore reduce the information leakage and, at the same time, increase the AN received by Eve. An FPA array, however, can hardly achieve this decorrelation when Eve is located in a direction close to that of an LU. If $\kappa_{\mathrm{e}} = 0$, then $\tilde{R}_{\mathrm{e},m} = \log_2(1 + p_m/(q(N - M) + \bar{\sigma}_{\mathrm{e}}^2))$ does not depend on $\mathbf{t}$, since the statistical CSI of Eve carries no spatial information, and only the first mechanism remains.

Remark 2 (Single LU): If $M = 1$, then $\mathbf{\Phi}(\mathbf{t}) = N(\kappa_1 + 1)$ is a scalar, and $\eta_1 = \zeta_1(N-1)(\kappa_1+1)$ does not depend on $\mathbf{t}$.[3] In contrast,

$$c_1(\mathbf{t}) = \frac{\kappa_1 |\bar{\mathbf{h}}_1^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2/N + 1}{\kappa_1 + 1}, \tag{27}$$

$$d(\mathbf{t}) = \frac{(N-1)(\kappa_1+1)}{(N-1)(\kappa_1+1)+1}\Big(N + \frac{1}{\kappa_1+1} - c_1(\mathbf{t})\Big) \tag{28}$$

depend on $\mathbf{t}$ through the LoS correlation $|\bar{\mathbf{h}}_1^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ between the LU and Eve. Therefore, unlike in the rate-oriented MA design, where the MAs bring no gain for a single user [16, Remark 2], the MAs enhance the secrecy of a single LU by reducing this correlation. For $M = 1$, $\bar{\mathbf{w}}_1\bar{\mathbf{w}}_1^{\mathrm{H}} + \mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{I}_N$ further implies $\bar{X}_1 + \bar{Y} = N$, so that reducing the leakage is equivalent to increasing the AN received by Eve. This identity is nearly preserved by (27) and (28), since $c_1(\mathbf{t}) + d(\mathbf{t}) - N = (c_1(\mathbf{t}) - 1)/((N-1)(\kappa_1+1)+1) \in [-1/4, 1/4]$.

Remark 3 (High-SNR Behavior and AN Saving): Consider the information–AN power splitting with equal power among the LUs, where $p_m = \alpha P_{\mathrm{tot}}/M$ and $q = (1-\alpha)P_{\mathrm{tot}}/(N-M)$ with the power-splitting factor $\alpha \in (0,1)$. As $P_{\mathrm{tot}} \to \infty$, i.e., in the high signal-to-noise ratio (SNR) regime, the approximate secrecy rate of the m-th LU satisfies

$$\tilde{R}_m - \tilde{R}_{\mathrm{e},m} = \log_2 \frac{\alpha P_{\mathrm{tot}}\eta_m(\mathbf{t})}{M} - \log_2\Big(1 + \frac{\alpha\Theta_m(\mathbf{t})}{1-\alpha}\Big) + o(1), \tag{29}$$

where $o(1)$ vanishes as $P_{\mathrm{tot}} \to \infty$, and $\Theta_m(\mathbf{t}) \triangleq \frac{(N-M)\xi_m(\mathbf{t})}{M\psi(\mathbf{t})}$ is the ratio of the normalized information leakage to the normalized AN power received by Eve. The high-SNR slope of the approximate ESSR is thus M regardless of $\mathbf{t}$, and the antenna positions affect only the constant term. The value of α that maximizes the sum of the constant terms in (29) satisfies $\sum_{m=1}^{M}\big[\frac{1}{\alpha} - \frac{1}{1-\alpha} - \frac{\Theta_m - 1}{1+\alpha(\Theta_m-1)}\big] = 0$, and its solution for $\Theta_m = \Theta$, $\forall m \in \mathcal{M}$, is given by

$$\alpha^\star = \frac{1}{1+\sqrt{\Theta}}. \tag{30}$$

Since the MAs can decrease $c_m(\mathbf{t})$ and increase $d(\mathbf{t})$, they can decrease Θ and thus increase $\alpha^\star$, which reduces the required AN power. We refer to this effect as the AN-saving effect.

Replacing the ergodic secrecy rates in (P2) with (26) yields the following deterministic approximate problem:

$$(\mathrm{P3}):\ \max_{\mathbf{t},\mathbf{p},q}\ F(\mathbf{t},\mathbf{p},q) \triangleq \sum_{m=1}^{M}\left[\log_2\big(1+\eta_m(\mathbf{t})p_m\big) - \log_2\left(1 + \frac{\xi_m(\mathbf{t})p_m}{q\psi(\mathbf{t}) + \bar{\sigma}_{\mathrm{e}}^2}\right)\right] \tag{31a}$$

$$\text{s.t.}\quad \text{(14b)–(14e)},$$

where the operator $[\cdot]^+$ in (26) is dropped without loss of optimality: for any feasible $(\mathbf{t},\mathbf{p},q)$, setting $p_m = 0$ for every LU whose term in (31a) is negative yields another feasible solution whose objective value equals the approximate ESSR $\sum_{m=1}^{M}\tilde{R}_m^{\mathrm{sec}}(\mathbf{t},p_m,q)$ of the original solution, because the m-th term depends only on p_m and q for a given $\mathbf{t}$. Consequently, (P3) has the same optimal value as its counterpart with $[\cdot]^+$, every term in (31a) is nonnegative at any optimal solution of (P3), and $F(\mathbf{t},\mathbf{p},q) \le \sum_{m=1}^{M}\tilde{R}_m^{\mathrm{sec}}(\mathbf{t},p_m,q)$ holds for any feasible solution. Under equal power allocation among the LUs, however, the operator $[\cdot]^+$ must be retained.

[3]For $M = 1$, even the exact ergodic rate $\mathbb{E}\{\log_2(1 + p_1\|\mathbf{h}_1(\mathbf{t})\|^2/\sigma_1^2)\}$ does not depend on $\mathbf{t}$, because the distribution of $\|\mathbf{h}_1(\mathbf{t})\|^2$ depends on $\bar{\mathbf{h}}_1(\mathbf{t})$ only through $\|\bar{\mathbf{h}}_1(\mathbf{t})\|^2 = N$.

IV. Proposed Two-Timescale Optimization Algorithm

Problem (P3) is still nonconvex, because F is nonconcave with respect to $\mathbf{t}$ and the minimum-distance constraints in (14e) are nonconvex. In addition, $\mathbf{t}$, $\mathbf{p}$, and q are coupled in the objective function, and $\mathbf{p}$ and q are further coupled by the power constraint (14b). Nevertheless, (P3) has two structures that simplify its optimization. First, for given $\mathbf{t}$, the coefficients η_m, ξ_m, and ψ in (31a) are constants, and the optimal power allocation among the LUs for a given q can be derived in semi-closed form. Second, for given $\mathbf{p}$ and q, each antenna position affects F only through a rank-one update of $\mathbf{\Phi}(\mathbf{t})$ and $\mathbf{v}(\mathbf{t})$, so that F can be evaluated at many candidate positions at low cost. Accordingly, we alternately optimize the power allocation and the antenna positions in Sections IV-A and IV-B, respectively.

A. Power Allocation Optimization for Given $\mathbf{t}$

For given $\mathbf{t}$, (P3) reduces to

$$(\mathrm{P3.1}):\ \max_{\mathbf{p},q}\ \sum_{m=1}^{M} f_m(p_m;q) \tag{32a}$$

$$\text{s.t.}\quad \text{(14b)},\ \text{(14c)},$$

where

$$f_m(p;q) \triangleq \log_2(1+\eta_m p) - \log_2\big(1+\eta_{\mathrm{e},m}(q)\,p\big), \tag{33}$$

and $\eta_{\mathrm{e},m}(q) \triangleq \xi_m/(q\psi + \bar{\sigma}_{\mathrm{e}}^2)$, which decreases with q, is the effective channel gain of Eve for s_m. For brevity, the argument $\mathbf{t}$ is omitted in this subsection. Problem (P3.1) is nonconvex in general, because f_m is not jointly concave in p_m and q. For a given q, however, the optimal $\mathbf{p}$ can be obtained in semi-closed form from the first-order and second-order derivatives of f_m with respect to p, which are given by

$$\frac{\partial f_m}{\partial p} = \frac{\eta_m - \eta_{\mathrm{e},m}}{\ln 2\,(1+\eta_m p)(1+\eta_{\mathrm{e},m}p)}, \tag{34}$$

$$\frac{\partial^2 f_m}{\partial p^2} = \frac{1}{\ln 2}\left[\frac{\eta_{\mathrm{e},m}^2}{(1+\eta_{\mathrm{e},m}p)^2} - \frac{\eta_m^2}{(1+\eta_m p)^2}\right], \tag{35}$$

where the argument q of $\eta_{\mathrm{e},m}$ is omitted. If $\eta_m \le \eta_{\mathrm{e},m}$, then $f_m(p;q) \le 0$ for all $p \ge 0$, and thus $p_m = 0$ is optimal. If $\eta_m > \eta_{\mathrm{e},m}$, then $\eta_m(1+\eta_{\mathrm{e},m}p) - \eta_{\mathrm{e},m}(1+\eta_m p) = \eta_m - \eta_{\mathrm{e},m} > 0$, i.e., $\frac{\eta_m}{1+\eta_m p} > \frac{\eta_{\mathrm{e},m}}{1+\eta_{\mathrm{e},m}p}$, and (34) and (35) show that f_m is strictly increasing and strictly concave in $p \ge 0$. Let $\mathcal{M}_{\mathrm{a}}(q) \triangleq \{m \in \mathcal{M} \,|\, \eta_m > \eta_{\mathrm{e},m}(q)\}$ denote the set of LUs that can achieve positive secrecy rates, and $P_{\mathrm{s}}(q) \triangleq P_{\mathrm{tot}} - (N-M)q$ the power available for the confidential signals. The optimal $\mathbf{p}$ for a given q is then given by the following proposition.

Algorithm 1 Power Allocation for Given $\mathbf{t}$

Input: $\{\eta_m, \xi_m\}_{m\in\mathcal{M}}$, ψ, $\bar{\sigma}_{\mathrm{e}}^2$, P_{tot}, and the grid size Q.
Output: $q^\star$ and $\mathbf{p}^\star$.
1: **for** each q in a uniform grid of Q points over $[0, q_{\max}]$ **do**
2: Obtain $\mathbf{p}^\star$ by Proposition 2, where ν is found by bisection.
3: Evaluate $F^\star(q)$.
4: **end for**
5: Repeat Steps 1–4 over a uniform grid of Q points between the two neighbors of the best grid point.
6: **return** the best q as $q^\star$ and its optimal power allocation as $\mathbf{p}^\star$.

Proposition 2 (Secrecy Water-Filling): For a given q with $P_{\mathrm{s}}(q) > 0$ and $\mathcal{M}_{\mathrm{a}}(q) \neq \emptyset$, the optimal power allocation is $p_m^\star = 0$ for $m \notin \mathcal{M}_{\mathrm{a}}(q)$ and

$$p_m^\star(\nu) = \left[\frac{2(\rho_m - 1)}{\sqrt{(\eta_m - \eta_{\mathrm{e},m})^2 + 4\eta_m\eta_{\mathrm{e},m}\rho_m} + \eta_m + \eta_{\mathrm{e},m}}\right]^+ \tag{36}$$

for $m \in \mathcal{M}_{\mathrm{a}}(q)$, where $\rho_m = \rho_m(\nu) \triangleq (\eta_m - \eta_{\mathrm{e},m})/(\nu \ln 2)$, and the water level $\nu > 0$ is the unique solution of $\sum_{m\in\mathcal{M}_{\mathrm{a}}(q)} p_m^\star(\nu) = P_{\mathrm{s}}(q)$.

Proof: Since the objective is strictly concave and increasing in $\{p_m\}_{m\in\mathcal{M}_{\mathrm{a}}(q)}$ and Slater's condition holds, the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for optimality, and the power constraint is active at the optimum [24]. Let $\nu > 0$ denote the dual variable associated with the power constraint. The KKT conditions yield $\partial f_m/\partial p = \nu$ if $p_m^\star > 0$ and $\partial f_m/\partial p|_{p=0} \leq \nu$ otherwise. From (34), the former is equivalent to the quadratic equation $(1+\eta_m p)(1+\eta_{\mathrm{e},m}p) = \rho_m(\nu)$ in p, whose positive root, written in a numerically stable form, is given by (36). The latter is equivalent to $\rho_m(\nu) \leq 1$, in which case (36) gives $p_m^\star = 0$. Finally, since $\sum_{m\in\mathcal{M}_{\mathrm{a}}(q)} p_m^\star(\nu)$ is continuous and strictly decreasing in ν over $(0, \max_{m\in\mathcal{M}_{\mathrm{a}}(q)}(\eta_m - \eta_{\mathrm{e},m})/\ln 2)$, ranging from $+\infty$ to 0, the water level is unique. ∎

Since $p_m^\star(\nu)$ is nonincreasing in ν, the water level in Proposition 2 can be found by bisection.

Remark 4 (Relation to Classic Water-Filling): As $\eta_{\mathrm{e},m} \to 0$, (36) reduces to the classic water-filling solution $[1/(\nu\ln 2) - 1/\eta_m]^+$. Equation (36) can therefore be regarded as a secrecy water-filling solution, which switches off the LUs whose effective channel gains do not exceed that of Eve.

Let $F^\star(q) \triangleq \sum_{m\in\mathcal{M}} f_m(p_m^\star; q)$ denote the optimal value of (P3.1) for a given q, with $F^\star(q) = 0$ if $\mathcal{M}_{\mathrm{a}}(q) = \emptyset$ or $P_{\mathrm{s}}(q) = 0$. The AN power per dimension is then optimized as

$$q^\star = \arg\max_{0\leq q\leq q_{\max}} F^\star(q), \tag{37}$$

where $q_{\max} \triangleq P_{\mathrm{tot}}/(N - M)$. Increasing q reduces the effective channel gains of Eve at the cost of less power for the confidential signals. Because $F^\star(q)$ is not necessarily unimodal, (37) is tackled by a coarse grid search followed by a finer one around the best grid point (Algorithm 1).

B. Antenna Position Optimization for Given $\mathbf{p}$ and q

For given $\mathbf{p}$ and q, (P3) reduces to

$$(\mathrm{P3.2}):\ \max_{\mathbf{t}}\ F(\mathbf{t}) \tag{38a}$$
$$\text{s.t.}\ \ \mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, \tag{38b}$$
$$\|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min},\ \forall n, i \in \mathcal{N},\ n \neq i, \tag{38c}$$

where $F(\mathbf{t}) \triangleq F(\mathbf{t}, \mathbf{p}, q)$. Problem (P3.2) is difficult to solve for two reasons. First, by Remark 1, $F(\mathbf{t})$ depends on $\mathbf{t}_n$ through the phases $\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}(\mathbf{a}_j - \mathbf{a}_i)$, $i, j \in \mathcal{M}\cup\{\mathrm{e}\}$, which rotate by 2π when $\mathbf{t}_n$ moves by $\lambda/\|\mathbf{a}_j - \mathbf{a}_i\|$ along $\mathbf{a}_j - \mathbf{a}_i$, so that $F(\mathbf{t})$ oscillates on the scale of a wavelength and has many local maxima within $\mathcal{C}$. Second, the minimum-distance constraints are nonconvex. To reduce the problem dimension, we optimize the antenna positions in an element-wise manner similar to [16]: $\mathbf{t}_n$ is updated with the other positions fixed, for $n = 1, \ldots, N$ in turn. Let $F(\mathbf{t}_n)$ denote F as a function of $\mathbf{t}_n$, with the other positions fixed at their latest values. The subproblem for the n-th MA is given by

$$(\mathrm{P3.2}.n):\ \max_{\mathbf{t}_n}\ F(\mathbf{t}_n) \tag{39a}$$
$$\text{s.t.}\ \ \mathbf{t}_n \in \mathcal{C}, \tag{39b}$$
$$\|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min},\ \forall i \in \mathcal{N}\setminus\{n\}. \tag{39c}$$

Although (P3.2.n) has only two real variables, $F(\mathbf{t}_n)$ is multimodal. A purely local method, e.g., the MM method adopted in [16], therefore tends to converge to a local maximum near the current position of the MA, so that the MAs move slowly across $\mathcal{C}$ and the AO iterations may stagnate at an inferior solution (see Section V-C). We thus tackle (P3.2.n) in two stages: Stage 1 selects the best candidate position by a grid search, and Stage 2 refines this position by the MM method. The cost of the grid search is kept low by the following rank-one structure.

1) Rank-one structure: Let $\boldsymbol{\ell}_n(\mathbf{t}_n) \in \mathbb{C}^{M\times 1}$ denote the conjugate transpose of the n-th row of $\mathbf{L}(\mathbf{t})$, i.e.,

$$\boldsymbol{\ell}_n(\mathbf{t}_n) = \mathbf{K}^{1/2}\left[e^{-j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_1}, \ldots, e^{-j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_M}\right]^{\mathrm{T}}. \tag{40}$$

Since $\mathbf{L}^{\mathrm{H}}\mathbf{L} = \sum_{i=1}^N \boldsymbol{\ell}_i\boldsymbol{\ell}_i^{\mathrm{H}}$ and $\mathbf{L}^{\mathrm{H}}\bar{\mathbf{g}} = \sum_{i=1}^N \boldsymbol{\ell}_i e^{j\frac{2\pi}{\lambda}\mathbf{t}_i^{\mathrm{T}}\mathbf{a}_{\mathrm{e}}}$, we have

$$\boldsymbol{\Phi}(\mathbf{t}) = \boldsymbol{\Gamma}_n + \boldsymbol{\ell}_n(\mathbf{t}_n)\boldsymbol{\ell}_n^{\mathrm{H}}(\mathbf{t}_n), \tag{41}$$
$$\mathbf{v}(\mathbf{t}) = \mathbf{r}_n + \boldsymbol{\ell}_n(\mathbf{t}_n)e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_{\mathrm{e}}}, \tag{42}$$

where $\boldsymbol{\Gamma}_n \triangleq \sum_{i\neq n}\boldsymbol{\ell}_i(\mathbf{t}_i)\boldsymbol{\ell}_i^{\mathrm{H}}(\mathbf{t}_i) + N\mathbf{I}_M$ and $\mathbf{r}_n \triangleq \sum_{i\neq n}\boldsymbol{\ell}_i(\mathbf{t}_i)e^{j\frac{2\pi}{\lambda}\mathbf{t}_i^{\mathrm{T}}\mathbf{a}_{\mathrm{e}}}$ do not depend on $\mathbf{t}_n$. By the Sherman–Morrison formula [23],

$$\boldsymbol{\Omega}(\mathbf{t}) = \boldsymbol{\Gamma}_n^{-1} - \frac{\boldsymbol{\Gamma}_n^{-1}\boldsymbol{\ell}_n(\mathbf{t}_n)\boldsymbol{\ell}_n^{\mathrm{H}}(\mathbf{t}_n)\boldsymbol{\Gamma}_n^{-1}}{1 + \boldsymbol{\ell}_n^{\mathrm{H}}(\mathbf{t}_n)\boldsymbol{\Gamma}_n^{-1}\boldsymbol{\ell}_n(\mathbf{t}_n)}. \tag{43}$$

Consequently, once $\boldsymbol{\Gamma}_n^{-1}$ and $\mathbf{r}_n$ are available, ω_m, c_m, d, and thus $F(\mathbf{t}_n)$ can be evaluated at any position with only $\mathcal{O}(M^2)$ operations.

2) Stage 1 (grid search): Let $\mathcal{G} \subset \mathcal{C}$ denote a uniform grid over $\mathcal{C}$, $\mathbf{t}_n^{\mathrm{cur}}$ the current position of the n-th MA, and $\mathcal{G}_n \triangleq \{\mathbf{t}_n \in \mathcal{G} \,|\, \|\mathbf{t}_n - \mathbf{t}_i\| \geq D_{\min},\ \forall i \neq n\}$ the set of grid points that are feasible for (P3.2.n). Using (43), we evaluate $F(\mathbf{t}_n)$ at all the candidates in $\mathcal{G}_n \cup \{\mathbf{t}_n^{\mathrm{cur}}\}$ and obtain

$$\hat{\mathbf{t}}_n = \arg\max_{\mathbf{t}_n\in\mathcal{G}_n\cup\{\mathbf{t}_n^{\mathrm{cur}}\}} F(\mathbf{t}_n), \tag{44}$$

which requires $\mathcal{O}(|\mathcal{G}|M^2)$ operations. Since the current position is included in the candidates, $F(\hat{\mathbf{t}}_n) \geq F(\mathbf{t}_n^{\mathrm{cur}})$. The grid spacing is set to a fraction of the shortest period $\lambda/\max_{i,j\in\mathcal{M}\cup\{\mathrm{e}\}}\|\mathbf{a}_j-\mathbf{a}_i\|$ of the above phases, which is at least $\lambda/2$ because $\|\mathbf{a}_i\|\leq 1$. Because $F(\mathbf{t}_n)$ is a rational function of these phase terms, the sufficiency of this spacing is not guaranteed analytically and is verified numerically in Section V-A.

3) Stage 2 (MM refinement): Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, we refine the position by the MM method [25], where $\mathbf{t}_n^{(k)}$ denotes the position of the n-th MA in the k-th MM iteration. To construct a surrogate function, we derive the gradient of $F(\mathbf{t}_n)$. Let $u_n \in \{x_n, y_n\}$ denote the coordinate of $\mathbf{t}_n$ along the axis $u \in \{x, y\}$, and define $\mathbf{\Lambda}_u \triangleq \mathrm{diag}(a_{1,u},\ldots,a_{M,u})$, where $a_{m,x} \triangleq \cos\theta_m \sin\phi_m$ and $a_{m,y} \triangleq \sin\theta_m$ are the entries of $\mathbf{a}_m$, and $a_{\mathrm{e},u}$ is the corresponding entry of $\mathbf{a}_{\mathrm{e}}$. From (40)–(42), the derivatives of $\boldsymbol{\ell}_n$, $\mathbf{v}$, and $\mathbf{\Omega}$ with respect to u_n are given by

$$\dot{\boldsymbol{\ell}}_n \triangleq \frac{\partial \boldsymbol{\ell}_n}{\partial u_n} = -j\frac{2\pi}{\lambda}\mathbf{\Lambda}_u \boldsymbol{\ell}_n, \tag{45}$$

$$\dot{\mathbf{v}} \triangleq \frac{\partial \mathbf{v}}{\partial u_n} = j\frac{2\pi}{\lambda}\big(a_{\mathrm{e},u}\mathbf{I}_M - \mathbf{\Lambda}_u\big)\boldsymbol{\ell}_n e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^{\mathrm{T}}\mathbf{a}_{\mathrm{e}}}, \tag{46}$$

$$\dot{\mathbf{\Omega}} \triangleq \frac{\partial \mathbf{\Omega}}{\partial u_n} = -\mathbf{\Omega}\big(\dot{\boldsymbol{\ell}}_n\boldsymbol{\ell}_n^{\mathrm{H}} + \boldsymbol{\ell}_n\dot{\boldsymbol{\ell}}_n^{\mathrm{H}}\big)\mathbf{\Omega}, \tag{47}$$

where (47) follows from $\partial\mathbf{\Phi}^{-1} = -\mathbf{\Phi}^{-1}(\partial\mathbf{\Phi})\mathbf{\Phi}^{-1}$. Using (45)–(47), the derivatives $\dot{\omega}_m \triangleq \partial\omega_m/\partial u_n$, $\dot{c}_m \triangleq \partial c_m/\partial u_n$, and $\dot{d} \triangleq \partial d/\partial u_n$ are derived in closed form in Appendix C. Then, the derivatives of the effective channel gains η_m and $\eta_{\mathrm{e},m} = \xi_m/(q\psi+\bar{\sigma}_{\mathrm{e}}^2)$ are given by $\dot{\eta}_m \triangleq \partial\eta_m/\partial u_n = -\eta_m\dot{\omega}_m/\omega_m$ and $\dot{\eta}_{\mathrm{e},m} \triangleq \partial\eta_{\mathrm{e},m}/\partial u_n = \kappa_{\mathrm{e}}\eta_{\mathrm{e},m}\big(\dot{c}_m/\xi_m - q\dot{d}/(q\psi+\bar{\sigma}_{\mathrm{e}}^2)\big)$, respectively. Since $F(\mathbf{t}_n) = \sum_{m=1}^{M}\big[\log_2(1+\eta_m p_m) - \log_2(1+\eta_{\mathrm{e},m}p_m)\big]$, the entries of the gradient $\nabla_{\mathbf{t}_n}F = [\partial F/\partial x_n,\ \partial F/\partial y_n]^{\mathrm{T}}$ are given by

$$\frac{\partial F}{\partial u_n} = \frac{1}{\ln 2}\sum_{m=1}^{M}\left[\frac{p_m\dot{\eta}_m}{1+\eta_m p_m} - \frac{p_m\dot{\eta}_{\mathrm{e},m}}{1+\eta_{\mathrm{e},m}p_m}\right]. \tag{48}$$

Since $F(\mathbf{t}_n)$ is twice continuously differentiable and $\mathcal{C}$ is compact, $\nabla_{\mathbf{t}_n}F$ is Lipschitz continuous over $\mathcal{C}$. By the descent lemma [26, Lemma 5.7], for any δ_n no smaller than this Lipschitz constant, the following quadratic function is a lower bound of $F(\mathbf{t}_n)$ that is tight at $\mathbf{t}_n^{(k)}$:

$$F(\mathbf{t}_n) \geq \hat{F}_n(\mathbf{t}_n|\mathbf{t}_n^{(k)}) \triangleq F(\mathbf{t}_n^{(k)}) + \nabla_{\mathbf{t}_n}F(\mathbf{t}_n^{(k)})^{\mathrm{T}}\big(\mathbf{t}_n - \mathbf{t}_n^{(k)}\big) - \frac{\delta_n}{2}\big\|\mathbf{t}_n - \mathbf{t}_n^{(k)}\big\|^2. \tag{49}$$

Because the Lipschitz constant is difficult to obtain, δ_n is determined by backtracking [26]: starting from $\max\{\delta_0, \delta_n^{\mathrm{pre}}/2\}$, where $\delta_0 > 0$ is a constant and δ_n^{pre} is the previously accepted value, δ_n is doubled until the update below satisfies $F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)}|\mathbf{t}_n^{(k)})$, which holds once δ_n reaches the Lipschitz constant.

For the minimum-distance constraints, $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ is convex with respect to $\mathbf{t}_n$ and is therefore lower-bounded by its first-order Taylor expansion at $\mathbf{t}_n^{(k)}$ [16], i.e.,

$$\|\mathbf{t}_n - \mathbf{t}_i\|^2 \geq \big(\mathbf{t}_n^{(k)} - \mathbf{t}_i\big)^{\mathrm{T}}\big(2\mathbf{t}_n - \mathbf{t}_n^{(k)} - \mathbf{t}_i\big). \tag{50}$$

Algorithm 2 Two-Timescale AO Algorithm for (P3)

Input: Statistical CSI, P_{tot}, $\mathcal{C}$, $D_{\min}$, Q, $\mathcal{G}$, δ_0, ϵ_{in}, N_{MM}, ϵ, N_{AO}.
Output: $\mathbf{t}^\star$, $\mathbf{p}^\star$, and $q^\star$.

1: Initialize feasible positions $\mathbf{t}$ and obtain $(\mathbf{p}, q)$ by Algorithm 1.
2: **repeat**
3: **for** $n = 1$ to N **do**
4: *Stage 1:* Compute $\mathbf{\Gamma}_n^{-1}$ and $\mathbf{r}_n$, and obtain $\hat{\mathbf{t}}_n$ by (44).
5: *Stage 2:* Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, update $\mathbf{t}_n$ by (52) with backtracking on δ_n until the increase of F normalized by $\max\{1,|F|\}$ is below ϵ_{in} or N_{MM} iterations are reached.
6: **end for**
7: Obtain $(\mathbf{p}', q')$ by Algorithm 1 for the updated $\mathbf{t}$, and set $(\mathbf{p}, q) \leftarrow (\mathbf{p}', q')$ if $F(\mathbf{t}, \mathbf{p}', q') \geq F(\mathbf{t}, \mathbf{p}, q)$.
8: **until** the increase of F normalized by $\max\{1,|F|\}$ is below ϵ or N_{AO} iterations are reached

Replacing $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ in the minimum-distance constraints with the right-hand side of (50) yields the linear constraints

$$\big(\mathbf{t}_n^{(k)} - \mathbf{t}_i\big)^{\mathrm{T}}\big(2\mathbf{t}_n - \mathbf{t}_n^{(k)} - \mathbf{t}_i\big) \geq D_{\min}^2, \quad \forall i \neq n, \tag{51}$$

whose feasible set is a convex subset of that of the minimum-distance constraints with respect to $\mathbf{t}_n$. Combining (49) and (51), we update $\mathbf{t}_n$ by solving

$$\mathbf{t}_n^{(k+1)} = \arg\max_{\mathbf{t}_n}\ \hat{F}_n(\mathbf{t}_n|\mathbf{t}_n^{(k)}) \quad \text{s.t. } \mathbf{t}_n \in \mathcal{C},\ (51). \tag{52}$$

Completing the square shows that (52) is equivalent to projecting $\mathbf{t}_n^{(k)} + \nabla_{\mathbf{t}_n}F(\mathbf{t}_n^{(k)})/\delta_n$ onto the convex polygon defined by $\mathcal{C}$ and (51). The solution is this point itself if it is feasible and otherwise lies on an edge or at a vertex of the polygon, so that enumerating the projections onto the $N+3$ constraint lines and their pairwise intersections yields the exact solution. Since $\mathbf{t}_n^{(k)}$ is feasible for (52), the update satisfies

$$F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)}|\mathbf{t}_n^{(k)}) \geq \hat{F}_n(\mathbf{t}_n^{(k)}|\mathbf{t}_n^{(k)}) = F(\mathbf{t}_n^{(k)}), \tag{53}$$

and $\mathbf{t}_n^{(k+1)}$ is feasible for (P3.2.n) due to (50). The MM iterations are repeated until the increase of F normalized by $\max\{1,|F|\}$ is below ϵ_{in} or N_{MM} iterations are reached.

C. Overall Algorithm, Convergence, and Complexity

Algorithm 2 summarizes the overall AO algorithm for (P3), whose convergence, initialization, and complexity are discussed below.

Convergence: In each iteration of Algorithm 2, the grid search in (44) does not decrease the objective value of (P3) because the current position is among the candidates, and neither do the MM updates in (52) due to (53). The power allocation is likewise accepted only if the objective value does not decrease, so the objective value is nondecreasing over the iterations. It is also upper-bounded by $\sum_{m=1}^{M}\log_2(1+\zeta_m(N-M)(\kappa_m+1)P_{\mathrm{tot}})$ according to (19) and $p_m \leq P_{\mathrm{tot}}$, and it is therefore guaranteed to converge. This guarantee concerns the objective value only; it does not imply that the iterates converge to a stationary point or a globally optimal solution of (P3).

Initialization: Since (P3) is nonconvex, the solution obtained by Algorithm 2 may depend on the initial positions. This dependence is weaker than for purely local updates, because

the grid search in Stage 1 allows each MA to move to any feasible grid point in $\mathcal{C}$ in every iteration. In addition, Algorithm 2 can be run from several initializations, e.g., the positions of the FPA array and random feasible positions, and the best solution is retained.

Complexity: In each AO iteration, forming $\mathbf{\Phi}$ and $\mathbf{\Omega}$ requires $\mathcal{O}(NM^2 + M^3)$ operations. For each MA, $\mathbf{\Gamma}_n^{-1}$ is obtained from $\mathbf{\Omega}$ by a rank-one update. Stage 1 requires $\mathcal{O}(|\mathcal{G}|(M^2 + N))$ operations, including the feasibility check of the grid points. In Stage 2, the gradient in (48) and each evaluation of F require $\mathcal{O}(M^2)$ operations based on (43), and solving (52) by enumeration requires $\mathcal{O}(N^3)$ operations. Algorithm 1 requires $\mathcal{O}(QN_{\mathrm{b}}M)$ operations, where N_{b} is the number of bisection iterations. The complexity of Algorithm 2 per initialization is therefore $\mathcal{O}\big(N_{\mathrm{AO}}\big(NM^2 + M^3 + N|\mathcal{G}|(M^2+N) + NN_{\mathrm{MM}}N_{\mathrm{bt}}(M^2+N^3) + QN_{\mathrm{b}}M\big)\big)$, where N_{bt} is the average number of backtracking trials. In contrast to stochastic optimization [18], this complexity does not depend on the number of channel realizations. Since Algorithm 2 depends only on the slowly varying statistical CSI, it is executed only once per frame, whereas the ZF precoding and AN generation in each CI require $\mathcal{O}(NM^2+M^3)$ operations.

V. Simulation Results

A. Simulation Setup and Benchmarks

Unless otherwise specified, we set $N = 8$, $M = 3$, $A = 4\lambda$, $D_{\min} = \lambda/2$, $\kappa_m = \kappa_{\mathrm{e}} = 10$ dB, $\beta_m = \beta_{\mathrm{e}} = 1$, and $\sigma_m^2 = \sigma_{\mathrm{e}}^2 = \sigma^2$, so that $P_{\mathrm{tot}}/\sigma^2$ is the transmit SNR, which is 20 dB by default. The elevation and azimuth AoDs of the LUs are independently and uniformly distributed within $[-\pi/3, \pi/3]$. Similar to [15], Eve is located in a direction close to that of LU 1, which represents the challenging case discussed in Section I. Its AoDs are $\theta_{\mathrm{e}} = \theta_1 + \Delta\cos\tilde{\phi}$ and $\phi_{\mathrm{e}} = \phi_1 + \Delta\sin\tilde{\phi}$, so that they deviate from those of LU 1 by the AoD offset Δ in the direction specified by $\tilde{\phi}$, which is uniformly distributed over $[0, 2\pi)$. Note that Δ is an offset in the elevation–azimuth coordinates rather than the physical angle between the two directions. By default, $\Delta = 0.1$ rad, for which the normalized LoS correlation $|\bar{\mathbf{h}}_1^{\mathrm{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|/N$ between Eve and LU 1 exceeds 0.94 for the FPA; Eve thus lies well within the main lobe of the beam steered toward LU 1. The case where the AoDs of Eve are independent of those of the LUs is considered in Section V-H.

For Algorithm 2, the grid $\mathcal{G}$ in Stage 1 has a spacing of $\lambda/10$, which is at most one fifth of the shortest phase period in Section IV-B and yields $|\mathcal{G}| = 1681$ for $A = 4\lambda$. Halving this spacing changes the average ESSR of the proposed design by only 0.01% over 20 AoD realizations. The remaining parameters are $\delta_0 = 1$, $\epsilon_{\mathrm{in}} = 10^{-6}$, $N_{\mathrm{MM}} = 20$, $\epsilon = 10^{-4}$, $N_{\mathrm{AO}} = 50$, $Q = 41$, and $N_{\mathrm{b}} = 60$. Algorithm 2 is executed from a $2\times(N/2)$ uniform planar array (UPA) with $\lambda/2$ spacing and two sets of random feasible positions, and the best solution is selected. All the results are averaged over 100 random AoD realizations, which are shared by all the schemes so that the comparisons are paired; the standard error of the average ESSR of the proposed design at 20 dB is 0.55% of its value. Since the closed-form approximations are used only for optimization, the ergodic rates of each design are evaluated with the exact SINRs in (12) and (13) over 5×10^3 channel realizations (10^4 in Fig. 3).

We compare the proposed design with four benchmarks: 1) **Rate-oriented MA**: The antenna positions maximize the approximate ergodic sum rate $\sum_{m=1}^{M}\log_2(1+\eta_m(\mathbf{t})P_{\mathrm{tot}}/M)$ of the LUs with equal power and ignore Eve [16]. For a fair comparison, this problem is tackled by the position update in Section IV-B with the same initializations. This benchmark isolates the gain of the secrecy-aware positioning enabled by (25). 2) **AS**: Similar to [15], the BS selects N of $2N$ candidate antennas, which form a UPA with 4 rows, $N/2$ columns, and $\lambda/2$ spacing. All the $\binom{2N}{N}$ subsets, e.g., 12870 for $N = 8$, are exhaustively ranked by the approximate ESSR in (P3) with the AN power searched over 21 grid points, and the power allocation of the selected subset is refined by Algorithm 1. AS requires no antenna movement but twice as many antennas and an RF switching network. 3) **Sparse FPA**: A $2\times(N/2)$ UPA spans $\mathcal{C}$ with spacings $2A/(N-2)$ and A along the x- and y-axes, respectively. This benchmark has the same aperture as the MAs and thus separates the gain of the optimized positions from that of the large aperture. 4) **FPA**: The BS adopts the $2\times(N/2)$ UPA with $\lambda/2$ spacing. All the schemes use the power allocation in Algorithm 1, so that their performance differences result only from the antenna positions.

B. Accuracy of the Closed-Form Approximations

Fig. 3 evaluates the accuracy of (18) and (25) at the designs obtained by Algorithm 2. The approximate wiretap rate in (25) agrees with the Monte Carlo results within maximum absolute errors of 0.052 and 0.013 bps/Hz over the SNRs in Fig. 3(a) and the Rician factors in Fig. 3(b), respectively, which indicates that it captures the LoS correlation between Eve and the LUs. By contrast, (18) underestimates the ergodic sum rate of the LUs, and the gap grows with the Rician factor. This underestimation stems from Jensen's inequality and from the central Wishart approximation in Section III-B, which is exact for Rayleigh fading but overestimates $\mathbb{E}\{[(\mathbf{H}^{\mathrm{H}}(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}\}$ by a factor approaching $N/(N-M)$ as the LoS components dominate. As a result, the approximate ESSR in (26) is lower than the actual ESSR in all the tested cases, e.g., 18.98 versus 20.75 bps/Hz at 20 dB. Since this gap is similar for all the schemes, namely between 1.52 and 1.76 bps/Hz at 20 dB, and smaller than the gains of the proposed design, the approximation remains a suitable design objective.

C. Convergence Behavior of Algorithm 2

Fig. 4 shows the objective value of (P3) versus the number of iterations of Algorithm 2, which is run without the stopping criterion. In all the settings, the objective value increases monotonically, and the average curves reach 99% of their total increase within 15 iterations. This fast convergence results from the grid search in Stage 1, which moves each MA directly to the best feasible grid point instead of letting it traverse the movable region through local updates. With the MM updates only, the objective value increases much more

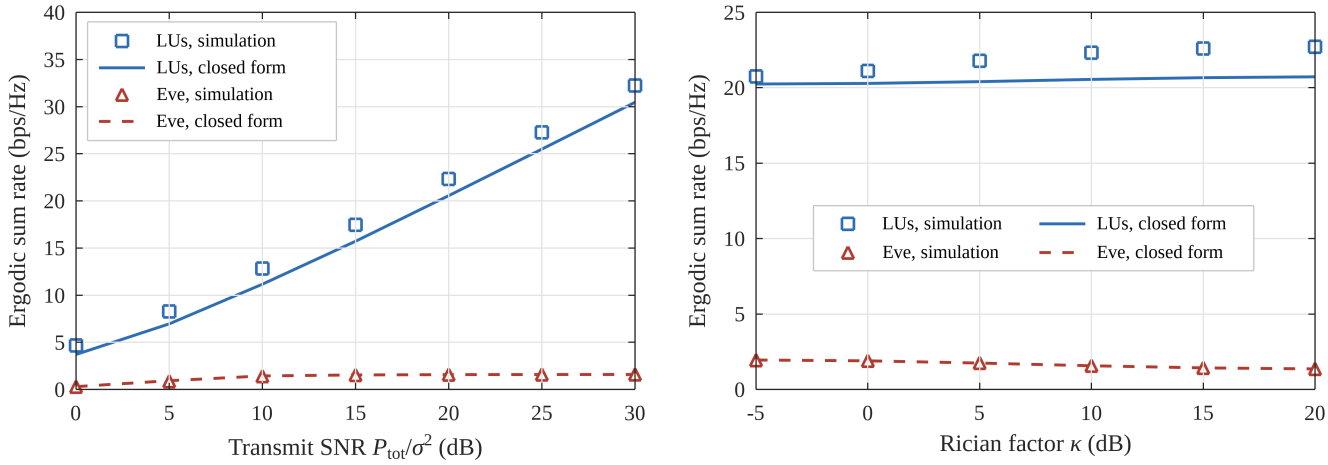


Fig. 3. Ergodic sum rates of the LUs and Eve for the proposed design by Monte Carlo simulations (markers) and by (18) and (25) (lines) versus (a) the transmit SNR P_{tot}/σ^2 and (b) the Rician factor $\kappa = \kappa_m = \kappa_{\text{e}}$.

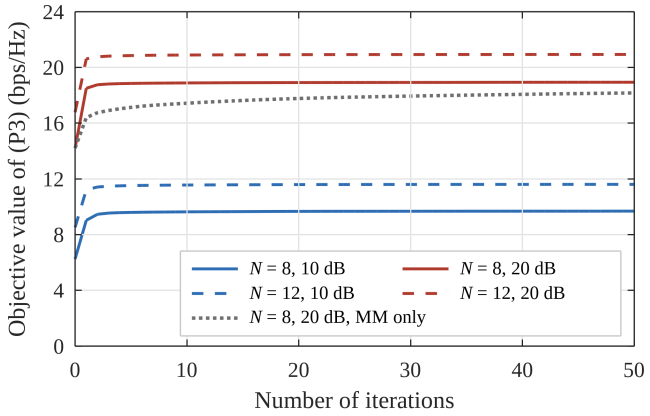


Fig. 4. Convergence behavior of Algorithm 2 for different numbers of MAs N and transmit SNRs P_{tot}/σ^2, averaged over 100 AoD realizations (UPA initialization). "MM only": without the grid search in Stage 1.

slowly and remains lower, because the MAs are trapped in the local maxima near their initial positions.

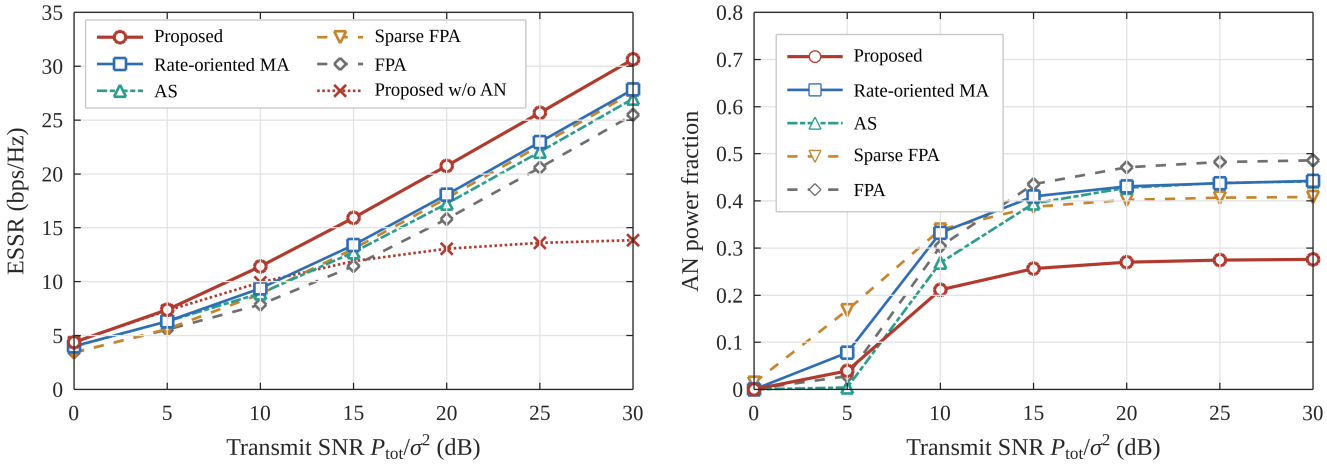
D. ESSR and AN Power Versus Transmit SNR

Fig. 5(a) shows that the proposed design achieves the highest ESSR over the entire SNR range, e.g., 14.8% and 31.2% higher than those of the rate-oriented MA and the FPA at 20 dB, respectively. Although the rate-oriented MA exploits the same movable region, its gain over the FPA is less than half of that of the proposed design, because its positions only decorrelate the LUs, whereas the proposed positions also reduce the information leakage and increase the AN received by Eve (see Remark 1). AS achieves an even smaller gain, since its $2N$ candidates form a compact array whose spatial resolution is insufficient to decorrelate the LoS components of Eve and LU 1. The sparse FPA has the same aperture as the MAs, yet it also achieves less than half of the gain of the proposed design over the FPA, because its fixed positions cannot adapt the LoS correlations to the AoDs of the LUs and Eve. The secrecy gain of the MAs therefore stems mainly from the optimized positions rather than from the large aperture. At high SNR, the absolute ESSR gains of the proposed design approach constants, since the antenna positions improve the constant term rather than the slope of the ESSR (see Remark 3). Without AN, i.e., with $q = 0$ in Algorithm 2, the ESSR of the proposed design saturates at high SNR, because the SINRs of the LUs and Eve then grow at the same rate with the transmit power [4].

Fig. 5(b) shows that the optimized AN power fraction increases with the SNR and saturates, as predicted by (30). The proposed design allocates much less power to the AN than all the benchmarks, e.g., an AN power fraction of 0.27 versus 0.47 for the FPA at 20 dB, whereas the AN power fractions of the other benchmarks are comparable to that of the FPA. This difference is consistent with the AN-saving effect in Remark 3: because the proposed positions reduce the information leakage and increase the AN received by Eve, less AN suffices, and more power remains for the confidential signals.

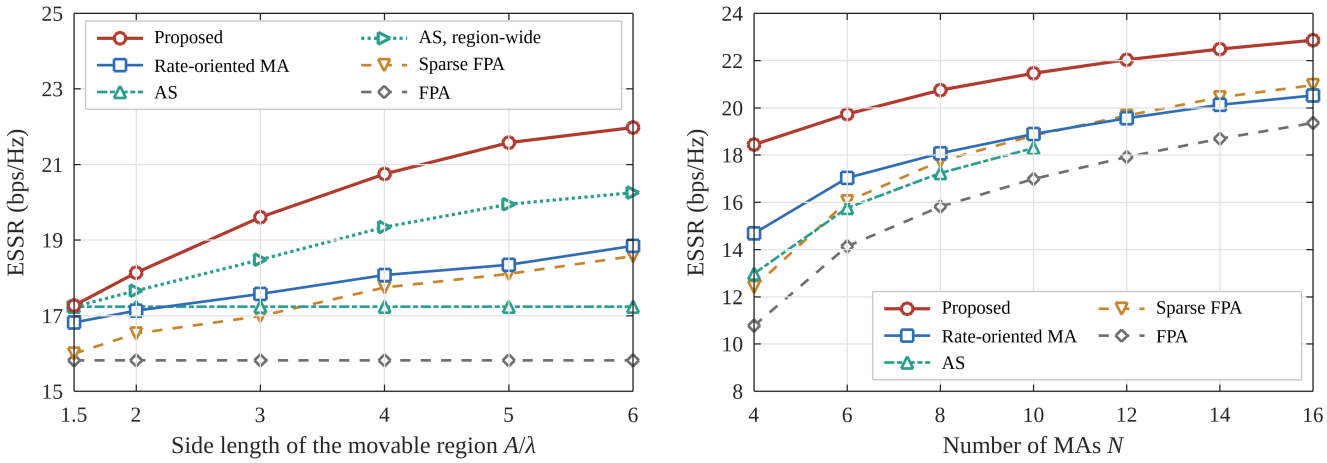
E. Impact of the Region Size and the Number of MAs

Fig. 6(a) shows the ESSR versus the side length A of the movable region. Here, we also consider AS with region-wide candidates, which form a 4×4 UPA spanning $\mathcal{C}$, whereas the FPA and AS with the compact candidate array do not depend on A. The gain of the proposed design grows with A, because a larger region provides more spatial degrees of freedom for decorrelating the LoS components of Eve and the LUs. The ESSRs of the rate-oriented MA and the sparse FPA increase much more slowly: the former exploits the enlarged region only to decorrelate the LUs, and the latter cannot adapt its positions to the AoDs. AS with region-wide candidates also benefits from the enlarged aperture and its Eve-aware selection metric, but the proposed design outperforms it with half the number of antennas, because the MAs can be placed



(a) (b)

Fig. 5. (a) ESSR and (b) optimized AN power fraction $(N-M)q/P_{\text{tot}}$ versus the transmit SNR P_{tot}/σ^2.



(a) (b)

Fig. 6. ESSR versus (a) the side length A of the movable region and (b) the number of MAs N.

anywhere in $\mathcal{C}$ rather than at fixed candidate positions. AS with the compact candidate array performs comparably to the proposed design only for the smallest region, which this array covers entirely.

Fig. 6(b) shows the ESSR versus the number of MAs N, where AS is shown only for $N \leq 10$ because the number of subsets in its exhaustive search grows exponentially with N. All the schemes benefit from more antennas, which provide a larger array gain and more dimensions for the AN. The gain of the proposed design over the FPA decreases with N, because the aperture of the FPA grows with N and allows it to separate Eve from LU 1 better even without position optimization. Nevertheless, the proposed design with $N = 4$ MAs outperforms the FPA with $N = 12$ antennas, and that with $N = 8$ MAs outperforms the rate-oriented MA with $N = 16$ MAs. The secrecy-aware positioning can thus substantially reduce the number of antennas and RF chains required for a target ESSR. The sparse FPA, whose grating lobes are strong for small N, outperforms the rate-oriented MA for $N \geq 14$, because more antennas in $\mathcal{C}$ suppress these lobes.

F. Impact of the AoD Offset

Fig. 7(a) shows the ESSR versus the AoD offset Δ for $M = 3$ and $M = 1$. For $M = 3$, all the schemes benefit from a larger Δ, because Eve becomes more separable from LU 1. When Eve is almost aligned with LU 1, the proposed design outperforms the rate-oriented MA only slightly, since even an aperture of 4λ can hardly decorrelate their LoS components. As Δ increases, the gain of the proposed design first increases and then decreases, because Eve eventually becomes separable even with Eve-unaware positions. For $M = 1$, the rate-oriented MA keeps its UPA initialization and thus coincides with the FPA, because its objective does not depend on the antenna positions (see Remark 2). The proposed design, by contrast, substantially improves the ergodic secrecy rate by reducing the LoS correlation $|\bar{\mathbf{h}}_1^{\text{H}}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ in (27) and (28), which confirms that the MAs enhance secrecy even for a single LU. The sparse FPA also benefits from its large aperture for

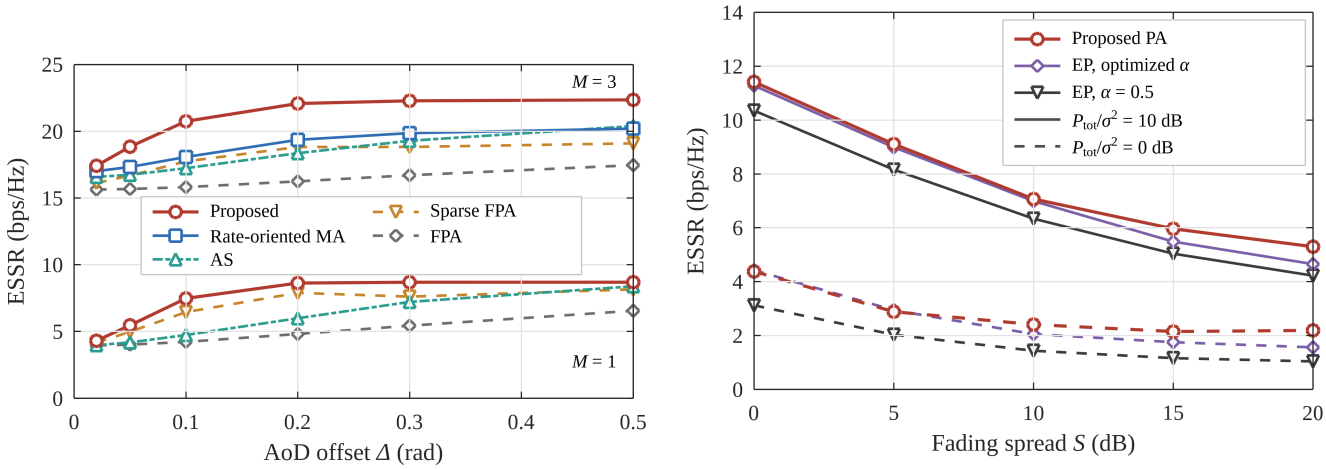


Fig. 7. (a) ESSR versus the AoD offset Δ between Eve and LU 1 for $M = 3$ and $M = 1$. (b) ESSR of the proposed power allocation (PA) and two equal-power (EP) benchmarks versus the fading spread S.

$M = 1$, but the proposed design achieves a further gain by adapting the positions to the AoDs.

G. Benefit of the Long-Term Power Allocation

Fig. 7(b) evaluates the proposed long-term power allocation, which combines the secrecy water-filling in Proposition 2 with the optimized AN power, versus the fading spread S, where the large-scale fading of the LUs is $[\beta_1, \beta_2, \beta_3] = [0, -S/2, -S]$ dB. Two equal-power (EP) benchmarks with $p_m = \alpha P_{\text{tot}}/M$ and $q = (1-\alpha)P_{\text{tot}}/(N-M)$ serve as references, where α is either optimized by a one-dimensional search [5] or fixed to 0.5 [15]. For all the schemes, the antenna positions are optimized by Algorithm 2. For the EP benchmarks, the objective retains $[\cdot]^+$ (see Section III-D), and its gradient in Stage 2 is computed over the LUs with positive terms, whose sum minorizes the objective.

EP with optimized α performs comparably to the proposed power allocation for small fading spreads, because the LUs then have similar effective channel gains, which the optimized positions balance further. At 0 dB and $S \leq 5$ dB, it is even slightly better, since the proposed allocation is optimal for the approximate objective rather than for the actual ESSR. As S increases, the gain of the proposed power allocation grows, since it switches off or reduces the power of the LUs whose effective channel gains are comparable to that of Eve (see Proposition 2). This gain decreases with the SNR, e.g., from 40% at 0 dB to 14% at 10 dB and to below 1% at 20 dB for $S = 20$ dB (the 20-dB values are not shown in Fig. 7(b)), because all the LUs achieve positive secrecy rates at high SNR, where nearly equal powers become optimal, as in classic water-filling. EP with $\alpha = 0.5$ performs worst in all the cases, especially at 0 dB, where little or no AN is needed. Both components of the long-term power allocation therefore matter: the information–AN power splitting improves the ESSR in all cases, and the per-user power allocation further benefits heterogeneous LUs at low-to-moderate SNR.

H. Robustness of the Secrecy Gain

Finally, we examine whether the gain of the proposed design relies on the worst-case eavesdropper model, the accurate statistical CSI of Eve, or the location of Eve close to an LU. First, the designs are evaluated against an Eve that treats the signals of the other LUs as interference (TIN), whose SINR has the additional term $\sum_{i \neq m} p_i |\mathbf{g}^{\text{H}}(\mathbf{t})\bar{\mathbf{w}}_i|^2$ in the denominator of (13). For $P_{\text{tot}}/\sigma^2 \in \{10, 20, 30\}$ dB, the ESSRs of all the schemes with AN increase by at most 1.53 bps/Hz, and the proposed design still achieves the highest ESSR, so its gain is not an artifact of the worst-case model. The proposed design also benefits the least from the TIN Eve, e.g., by 0.5 versus 1.5 bps/Hz for the rate-oriented MA at 20 dB, because its positions reduce the leakage of all the LUs and thus also the inter-user interference received by Eve. Without AN, however, its ESSR against the TIN Eve keeps increasing with the SNR, e.g., to 27.1 bps/Hz at 30 dB, because the inter-user interference then acts as AN, whereas it saturates at about 14 bps/Hz under the worst-case model (see Fig. 5(a)). The AN is therefore indispensable for a secrecy guarantee that does not rely on the decoding capability of Eve.

Second, all the schemes are designed with AoDs of Eve that deviate from the true ones by Δ_{err} in a random direction, as for Δ in Section V-A, and are evaluated with the true AoDs. The ESSRs of the benchmarks decrease by less than 2%, because only their power allocation and the antenna subset of AS depend on the AoDs of Eve. The ESSR of the proposed design decreases with Δ_{err}, since its positions are optimized for the assumed AoDs of Eve. At 20 dB, however, it decreases by only 3% for $\Delta_{\text{err}} = 0.05$ rad and still exceeds those of all benchmarks for $\Delta_{\text{err}} = \Delta = 0.1$ rad (18.73 versus at most 17.93 bps/Hz). This robustness is due to the smooth variation of the LoS correlations with the AoDs of Eve.

Third, when the AoDs of Eve are independent of those of the LUs and follow the same distribution, the proposed design still outperforms all the benchmarks, e.g., by 10% and 18%

compared with the rate-oriented MA and the sparse FPA for $M = 3$, whereas AS performs close to the proposed design for $M = 1$. Eve is then mostly well separated from the LUs, so that the compact candidates of AS can already decorrelate it from a single LU, whereas the large aperture of the MAs is essential when Eve is close to an LU.

VI. Conclusion

This paper investigated two-timescale secure transmission for an MA-enabled multiuser downlink with a passive Eve. The closed-form approximation of the ergodic wiretap rate revealed that, beyond decorrelating the LUs, the MAs enhance secrecy by reducing the information leakage and increasing the AN received by Eve. Building on this approximation, we developed an AO algorithm that combines secrecy water-filling with a two-stage position update and whose objective value is guaranteed to converge. Simulation results demonstrated that the proposed secrecy-aware positioning outperforms the rate-oriented MA design, AS, and FPA arrays, also for a single LU, while reducing the AN power and the number of antennas required for secure transmission. Since the present design assumes a single-antenna Eve and perfect CSI of the LUs, its extension to multiple multi-antenna eavesdroppers and imperfect CSI is left for future work.

Appendix A
Proof of Lemma 1

Let $\mathbf{U} \in \mathbb{C}^{N\times(N-M)}$ have orthonormal columns that span the null space of $\mathbf{H}^{\mathrm{H}}(\mathbf{t})$. Then, $\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{U}\mathbf{U}^{\mathrm{H}}$ and $Y = \|\mathbf{U}^{\mathrm{H}}\mathbf{g}(\mathbf{t})\|^2$. Since $\bar{\mathbf{w}}_m$ is a unit-norm vector in the column space of $\mathbf{H}(\mathbf{t})$, $\tilde{\mathbf{U}} \triangleq [\bar{\mathbf{w}}_m, \mathbf{U}]$ has orthonormal columns. Because $\tilde{\mathbf{g}}$ is independent of $\mathbf{H}(\mathbf{t})$, (3) yields $\mathbf{g}(\mathbf{t})|\mathbf{H}(\mathbf{t}) \sim \mathcal{CN}(\sqrt{\kappa_{\mathrm{e}}\tilde{\beta}_{\mathrm{e}}}\,\bar{\mathbf{g}}(\mathbf{t}), \tilde{\beta}_{\mathrm{e}}\mathbf{I}_N)$. Consequently, $\tilde{\mathbf{U}}^{\mathrm{H}}\mathbf{g}(\mathbf{t})$ is conditionally Gaussian with covariance matrix $\tilde{\beta}_{\mathrm{e}}\tilde{\mathbf{U}}^{\mathrm{H}}\tilde{\mathbf{U}} = \tilde{\beta}_{\mathrm{e}}\mathbf{I}_{N-M+1}$, so that its entries are conditionally independent. As a result, $X_m = |\bar{\mathbf{w}}_m^{\mathrm{H}}\mathbf{g}(\mathbf{t})|^2$ and $Y = \|\mathbf{U}^{\mathrm{H}}\mathbf{g}(\mathbf{t})\|^2$ are conditionally independent. Finally, (20) and (21) follow from applying $\mathbb{E}\{|z|^2\} = |\mu|^2 + \tilde{\beta}_{\mathrm{e}}$ and $\mathrm{Var}\{|z|^2\} = \tilde{\beta}_{\mathrm{e}}^2 + 2|\mu|^2\tilde{\beta}_{\mathrm{e}}$ for $z \sim \mathcal{CN}(\mu, \tilde{\beta}_{\mathrm{e}})$ to the entries of $\tilde{\mathbf{U}}^{\mathrm{H}}\mathbf{g}(\mathbf{t})$, together with $|\bar{\mathbf{w}}_m^{\mathrm{H}}\bar{\mathbf{g}}(\mathbf{t})|^2 = \bar{X}_m$ and $\|\mathbf{U}^{\mathrm{H}}\bar{\mathbf{g}}(\mathbf{t})\|^2 = \bar{Y}$.

Appendix B
Derivation of (23) and (24) and Proof of Proposition 1

For brevity, the argument $\mathbf{t}$ is omitted. Due to the scaling invariance noted in Section III-C, (15) yields $\bar{\mathbf{w}}_m = \breve{\mathbf{H}}(\breve{\mathbf{H}}^{\mathrm{H}}\breve{\mathbf{H}})^{-1}\mathbf{e}_m / \|\breve{\mathbf{H}}(\breve{\mathbf{H}}^{\mathrm{H}}\breve{\mathbf{H}})^{-1}\mathbf{e}_m\|$ and $\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{I}_N - \breve{\mathbf{H}}(\breve{\mathbf{H}}^{\mathrm{H}}\breve{\mathbf{H}})^{-1}\breve{\mathbf{H}}^{\mathrm{H}}$. We also use $\mathbb{E}\{\breve{\mathbf{H}}^{\mathrm{H}}\breve{\mathbf{H}}\} = \mathbf{\Phi}$, $\mathbb{E}\{\tilde{\mathbf{H}}\} = \mathbf{0}$, and $\mathbb{E}\{\tilde{\mathbf{H}}\mathbf{x}\mathbf{x}^{\mathrm{H}}\tilde{\mathbf{H}}^{\mathrm{H}}\} = \|\mathbf{x}\|^2\mathbf{I}_N$ for any deterministic $\mathbf{x}$.

Derivation of (23)*:* Replacing $(\breve{\mathbf{H}}^{\mathrm{H}}\breve{\mathbf{H}})^{-1}$ in the unnormalized ZF direction by $\mathbf{\Omega} = \mathbf{\Phi}^{-1}$ yields $\bar{\mathbf{w}}_m \approx \breve{\mathbf{H}}\mathbf{\Omega}\mathbf{e}_m / \|\breve{\mathbf{H}}\mathbf{\Omega}\mathbf{e}_m\|$. Then, we approximate the mean of $|\bar{\mathbf{g}}^{\mathrm{H}}\breve{\mathbf{H}}\mathbf{\Omega}\mathbf{e}_m|^2 / \|\breve{\mathbf{H}}\mathbf{\Omega}\mathbf{e}_m\|^2$ by the ratio of the means of its numerator and denominator, as in [19, Lemma 1]. These are two separate approximations, since $\mathbb{E}\{(\breve{\mathbf{H}}^{\mathrm{H}}\breve{\mathbf{H}})^{-1}\} \neq \mathbf{\Omega}$ in general. Since $\breve{\mathbf{H}} = \mathbf{L} + \tilde{\mathbf{H}}$ and $\mathbf{L}^{\mathrm{H}}\bar{\mathbf{g}} = \mathbf{v}$, we have

$$\begin{aligned}\mathbb{E}\{|\bar{\mathbf{g}}^{\mathrm{H}}\breve{\mathbf{H}}\mathbf{\Omega}\mathbf{e}_m|^2\} &= |\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{L}\mathbf{\Omega}\mathbf{e}_m|^2 + \|\bar{\mathbf{g}}\|^2\|\mathbf{\Omega}\mathbf{e}_m\|^2 \\ &= \left|[\mathbf{\Omega}\mathbf{v}]_m\right|^2 + N[\mathbf{\Omega}^2]_{m,m}, \qquad (54)\\ \mathbb{E}\{\|\breve{\mathbf{H}}\mathbf{\Omega}\mathbf{e}_m\|^2\} &= \mathbf{e}_m^{\mathrm{T}}\mathbf{\Omega}\mathbf{\Phi}\mathbf{\Omega}\mathbf{e}_m = \omega_m, \qquad (55)\end{aligned}$$

whose ratio is given by (23).

Derivation of (24)*:* Replacing $(\breve{\mathbf{H}}^{\mathrm{H}}\breve{\mathbf{H}})^{-1}$ in $\mathbf{I}_N - \mathbf{P}^{\perp}(\mathbf{t})$ by $\mathbf{\Omega}$ and using $\mathbb{E}\{\tilde{\mathbf{H}}\mathbf{\Omega}\tilde{\mathbf{H}}^{\mathrm{H}}\} = \mathrm{tr}(\mathbf{\Omega})\mathbf{I}_N$ yield

$$\mathbb{E}\{\mathbf{I}_N - \mathbf{P}^{\perp}(\mathbf{t})\} \approx \mathbf{\Pi} + \mathrm{tr}(\mathbf{\Omega})\mathbf{I}_N, \qquad \mathbf{\Pi} \triangleq \mathbf{L}\mathbf{\Omega}\mathbf{L}^{\mathrm{H}}, \quad (56)$$

where $\mathbf{\Pi}$ and $\mathrm{tr}(\mathbf{\Omega})\mathbf{I}_N$ represent the parts of the signal subspace induced by the LoS and NLoS components, respectively. However, (56) lets the NLoS component capture the fraction $\mathrm{tr}(\mathbf{\Omega})$ of every direction, including those captured by the LoS component, so that $\bar{\mathbf{g}}^{\mathrm{H}}(\mathbf{I}_N - \mathbf{\Pi} - \mathrm{tr}(\mathbf{\Omega})\mathbf{I}_N)\bar{\mathbf{g}}$ may become negative when $\bar{\mathbf{g}}$ is closely aligned with the column space of $\mathbf{L}$. To avoid this double counting, we let the NLoS component capture a fraction $\chi \in [0,1]$ of the complement of $\mathbf{\Pi}$ only, i.e.,

$$\mathbb{E}\{\mathbf{I}_N - \mathbf{P}^{\perp}(\mathbf{t})\} \approx \mathbf{\Pi} + \chi(\mathbf{I}_N - \mathbf{\Pi}). \quad (57)$$

The fraction χ is determined by the identity $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$, which holds for every realization of $\breve{\mathbf{H}}$. Since $\mathrm{tr}(\mathbf{\Pi}) = \mathrm{tr}\big(\mathbf{\Omega}(\mathbf{\Phi} - N\mathbf{I}_M)\big) = M - N\mathrm{tr}(\mathbf{\Omega})$, taking the trace of (57) yields $M - N\mathrm{tr}(\mathbf{\Omega}) + \chi\big(N - M + N\mathrm{tr}(\mathbf{\Omega})\big) = M$, i.e., $\chi = N\mathrm{tr}(\mathbf{\Omega})/(N - M + N\mathrm{tr}(\mathbf{\Omega}))$. Therefore, $\mathbb{E}\{\mathbf{P}^{\perp}(\mathbf{t})\} \approx (1-\chi)(\mathbf{I}_N - \mathbf{\Pi})$, which yields (24) due to $\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{\Pi}\bar{\mathbf{g}} = \mathbf{v}^{\mathrm{H}}\mathbf{\Omega}\mathbf{v}$.

Proof of Proposition 1: For the range, the matrix inversion lemma yields $\mathbf{I}_N - \mathbf{\Pi} = \mathbf{I}_N - \mathbf{L}(\mathbf{L}^{\mathrm{H}}\mathbf{L} + N\mathbf{I}_M)^{-1}\mathbf{L}^{\mathrm{H}} = (\mathbf{I}_N + \mathbf{L}\mathbf{L}^{\mathrm{H}}/N)^{-1} \succ \mathbf{0}$. Together with $\mathbf{\Pi} \succeq \mathbf{0}$, this implies $\mathbf{0} \preceq \mathbf{\Pi} \prec \mathbf{I}_N$. Moreover, $\mathbf{\Phi} \succeq N\mathbf{I}_M$ implies $0 < \mathrm{tr}(\mathbf{\Omega}) \leq M/N$, i.e., $1 - \chi \in [1 - M/N, 1)$. Hence, $d = (1-\chi)\bar{\mathbf{g}}^{\mathrm{H}}(\mathbf{I}_N - \mathbf{\Pi})\bar{\mathbf{g}} \in [0, N]$ due to $\|\bar{\mathbf{g}}\|^2 = N$. For c_m, the Cauchy–Schwarz inequality yields $|[\mathbf{\Omega}\mathbf{v}]_m|^2 = |\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{L}\mathbf{\Omega}\mathbf{e}_m|^2 \leq N\mathbf{e}_m^{\mathrm{T}}\mathbf{\Omega}\mathbf{L}^{\mathrm{H}}\mathbf{L}\mathbf{\Omega}\mathbf{e}_m$, so that the numerator of (23) is at most $N\mathbf{e}_m^{\mathrm{T}}\mathbf{\Omega}(\mathbf{L}^{\mathrm{H}}\mathbf{L} + N\mathbf{I}_M)\mathbf{\Omega}\mathbf{e}_m = N\omega_m$ due to $N \geq 1$, i.e., $c_m \in [0, N]$. The exact counterparts also lie in $[0, N]$, since $\bar{X}_m, \bar{Y} \leq \|\bar{\mathbf{g}}\|^2$ for the unit-norm $\bar{\mathbf{w}}_m$ and the projector $\mathbf{P}^{\perp}(\mathbf{t})$.

If $\kappa_m = 0$, $\forall m \in \mathcal{M}$, then $\mathbf{L} = \mathbf{0}$, $\breve{\mathbf{H}} = \tilde{\mathbf{H}}$, $\mathbf{\Phi} = N\mathbf{I}_M$, and $\mathbf{\Pi} = \mathbf{0}$, so that (23) and (24) reduce to $c_m = 1$ and $d = N - M$. In this case, the distribution of $\tilde{\mathbf{H}}$ is invariant to left multiplication by any unitary matrix, which is inherited by $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$. Hence, $\bar{\mathbf{w}}_m$ is uniformly distributed on the unit sphere, which yields $\mathbb{E}\{\bar{X}_m\} = \|\bar{\mathbf{g}}\|^2/N = 1$, and $\mathbb{E}\{\mathbf{P}^{\perp}(\mathbf{t})\}$ commutes with every unitary matrix and thus equals $(1 - M/N)\mathbf{I}_N$ due to $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$, which yields $\mathbb{E}\{\bar{Y}\} = N - M$.

Finally, let $\mathbf{K} = \kappa\mathbf{K}_0$ for a fixed diagonal matrix $\mathbf{K}_0 \succ \mathbf{0}$, so that $\mathbf{L} = \sqrt{\kappa}\mathbf{L}_0$ with $\mathbf{L}_0 \triangleq \bar{\mathbf{H}}\mathbf{K}_0^{1/2}$, which has full column rank if the LoS components of the LUs are linearly independent. As $\kappa \to \infty$, $\breve{\mathbf{H}}/\sqrt{\kappa} \to \mathbf{L}_0$ almost surely. Since $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are invariant to the scaling of $\breve{\mathbf{H}}$ and continuous at full-column-rank matrices, and $\bar{X}_m$ and $\bar{Y}$ are bounded by N, the dominated convergence theorem yields $\mathbb{E}\{\bar{X}_m\} \to |\bar{\mathbf{g}}^{\mathrm{H}}\bar{\mathbf{w}}_m^{\infty}|^2$ and $\mathbb{E}\{\bar{Y}\} \to \bar{\mathbf{g}}^{\mathrm{H}}\mathbf{P}_{\infty}^{\perp}\bar{\mathbf{g}}$, where

$\bar{\mathbf{w}}_m^{\infty} \triangleq \mathbf{L}_0(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}\mathbf{e}_m/\|\mathbf{L}_0(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}\mathbf{e}_m\|$ and $\mathbf{P}_{\infty}^{\perp} \triangleq \mathbf{I}_N - \mathbf{L}_0(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}\mathbf{L}_0^{\mathrm{H}}$. On the other hand, $\kappa\boldsymbol{\Omega} \to (\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}$, $\boldsymbol{\Pi} \to \mathbf{I}_N - \mathbf{P}_{\infty}^{\perp}$, and $\mathrm{tr}(\boldsymbol{\Omega}) \to 0$. Hence, d converges to $\bar{\mathbf{g}}^{\mathrm{H}}\mathbf{P}_{\infty}^{\perp}\bar{\mathbf{g}}$, and c_m converges to $|[(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}\mathbf{L}_0^{\mathrm{H}}\bar{\mathbf{g}}]_m|^2/[(\mathbf{L}_0^{\mathrm{H}}\mathbf{L}_0)^{-1}]_{m,m} = |\bar{\mathbf{g}}^{\mathrm{H}}\bar{\mathbf{w}}_m^{\infty}|^2$, since $N[\boldsymbol{\Omega}^2]_{m,m}$ decays as $\mathcal{O}(\kappa^{-2})$ whereas ω_m decays as $\mathcal{O}(\kappa^{-1})$. This completes the proof.

APPENDIX C
DERIVATIVES OF ω_m, c_m, AND d

Let $\tau_m \triangleq [\boldsymbol{\Omega}^2]_{m,m}$ and $\tilde{\mathbf{v}} \triangleq \boldsymbol{\Omega}\mathbf{v}$. Since $\boldsymbol{\Omega}$ is Hermitian, (47) can be rewritten as $\dot{\boldsymbol{\Omega}} = -(\boldsymbol{\Omega}\dot{\boldsymbol{\ell}}_n)(\boldsymbol{\Omega}\boldsymbol{\ell}_n)^{\mathrm{H}} - (\boldsymbol{\Omega}\boldsymbol{\ell}_n)(\boldsymbol{\Omega}\dot{\boldsymbol{\ell}}_n)^{\mathrm{H}}$. Hence, the derivatives of ω_m, τ_m, and $\mathrm{tr}(\boldsymbol{\Omega})$ with respect to u_n are given by

$$\dot{\omega}_m = [\dot{\boldsymbol{\Omega}}]_{m,m} = -2\mathrm{Re}\big\{[\boldsymbol{\Omega}\dot{\boldsymbol{\ell}}_n]_m[\boldsymbol{\Omega}\boldsymbol{\ell}_n]_m^*\big\}, \tag{58}$$

$$\dot{\tau}_m = 2\mathrm{Re}\big\{[\boldsymbol{\Omega}\dot{\boldsymbol{\Omega}}]_{m,m}\big\} = -2\mathrm{Re}\big\{[\boldsymbol{\Omega}^2\dot{\boldsymbol{\ell}}_n]_m[\boldsymbol{\Omega}\boldsymbol{\ell}_n]_m^* + [\boldsymbol{\Omega}^2\boldsymbol{\ell}_n]_m[\boldsymbol{\Omega}\dot{\boldsymbol{\ell}}_n]_m^*\big\}, \tag{59}$$

$$\frac{\partial \mathrm{tr}(\boldsymbol{\Omega})}{\partial u_n} = \mathrm{tr}(\dot{\boldsymbol{\Omega}}) = -2\mathrm{Re}\big\{\boldsymbol{\ell}_n^{\mathrm{H}}\boldsymbol{\Omega}^2\dot{\boldsymbol{\ell}}_n\big\}, \tag{60}$$

where (59) follows from $[\dot{\boldsymbol{\Omega}}\boldsymbol{\Omega}]_{m,m} = [\boldsymbol{\Omega}\dot{\boldsymbol{\Omega}}]_{m,m}^*$. For c_m in (23), the derivative of $\tilde{\mathbf{v}} = \boldsymbol{\Omega}\mathbf{v}$ is

$$\dot{\tilde{\mathbf{v}}} = \dot{\boldsymbol{\Omega}}\mathbf{v} + \boldsymbol{\Omega}\dot{\mathbf{v}} = -\boldsymbol{\Omega}\dot{\boldsymbol{\ell}}_n\big(\boldsymbol{\ell}_n^{\mathrm{H}}\tilde{\mathbf{v}}\big) - \boldsymbol{\Omega}\boldsymbol{\ell}_n\big(\dot{\boldsymbol{\ell}}_n^{\mathrm{H}}\tilde{\mathbf{v}}\big) + \boldsymbol{\Omega}\dot{\mathbf{v}}, \tag{61}$$

and thus

$$\dot{c}_m = \frac{2\mathrm{Re}\big\{[\tilde{\mathbf{v}}]_m^*[\dot{\tilde{\mathbf{v}}}]_m\big\} + N\dot{\tau}_m - c_m\dot{\omega}_m}{\omega_m}. \tag{62}$$

For $d = (1-\chi)(N - \mathbf{v}^{\mathrm{H}}\boldsymbol{\Omega}\mathbf{v})$ in (24) with $1-\chi = (N-M)/(N - M + N\mathrm{tr}(\boldsymbol{\Omega}))$ as in Appendix B, we have

$$\frac{\partial(\mathbf{v}^{\mathrm{H}}\boldsymbol{\Omega}\mathbf{v})}{\partial u_n} = 2\mathrm{Re}\{\tilde{\mathbf{v}}^{\mathrm{H}}\dot{\mathbf{v}}\} - 2\mathrm{Re}\big\{(\tilde{\mathbf{v}}^{\mathrm{H}}\dot{\boldsymbol{\ell}}_n)(\boldsymbol{\ell}_n^{\mathrm{H}}\tilde{\mathbf{v}})\big\}, \tag{63}$$

$$\dot{d} = -(1-\chi)\bigg(\frac{\partial(\mathbf{v}^{\mathrm{H}}\boldsymbol{\Omega}\mathbf{v})}{\partial u_n} + \frac{N(N - \mathbf{v}^{\mathrm{H}}\boldsymbol{\Omega}\mathbf{v})}{N - M + N\mathrm{tr}(\boldsymbol{\Omega})}\,\frac{\partial \mathrm{tr}(\boldsymbol{\Omega})}{\partial u_n}\bigg), \tag{64}$$

where (63) follows from $\mathbf{v}^{\mathrm{H}}\dot{\boldsymbol{\Omega}}\mathbf{v} = -2\mathrm{Re}\{(\tilde{\mathbf{v}}^{\mathrm{H}}\dot{\boldsymbol{\ell}}_n)(\boldsymbol{\ell}_n^{\mathrm{H}}\tilde{\mathbf{v}})\}$. Substituting (58), (62), and (64) into (48) yields $\nabla_{\mathbf{t}_n}F$. All the quantities in (58)–(64) follow from matrix-vector products with $\boldsymbol{\Omega}$, requiring $\mathcal{O}(M^2)$ operations.

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